# MILITARY SPECIFICATION

## MODULES, STANDARD ELECTRONIC

## STANDARD SERIAL OUTPUT, DIGITAL, KEY CODE DQD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic standard serial output module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.329 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.4.1 Cooling requirements. The module shall be cooled with 27.6°C maximum temperature ambient air at a minimum of 12 fps or with an alternate temperature and velocity which will produce the equivalent cooling condition.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DQD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A. The cross sectional area of the entire fin shall not exceed 57 percent of the area designated in MIL-STD-1389, figure 6.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/212-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. This specification includes a set of unique input patterns (table VII) capable of detecting all nonredundant failures in the module logic networks. These test patterns are generated by assuming "stuck at H state" and "stuck at L state" in each of the basic logic functions and applying the necessary input patterns to detect these failures. A "stuck at H state" indicates a failure resulting in a steady high-level output. A "stuck at L state" indicates a failure resulting in a steady low-level output. Unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the test patterns submitted do not meet these criteria, the additional tests required shall be



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added to table VII. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at L state" or "stuck at H state", the vendor shall be required to submit to the acquiring activity and Standard Electronic Modules Program Quality Assurance Activity the proposed new set of test patterns for approval.

3.10 2 Input requirements. The following input voltage levels shall be used for test patterns.

Low level - Input voltage shall be 0.8 V ±1%.
High level - Input voltage shall be 2.0 V ±1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows (per figure 16 output loads).

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum, $V_{CC}$ maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4 2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4 3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4 4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V and X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 5

4 5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 7.

4 5 2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 8 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient.



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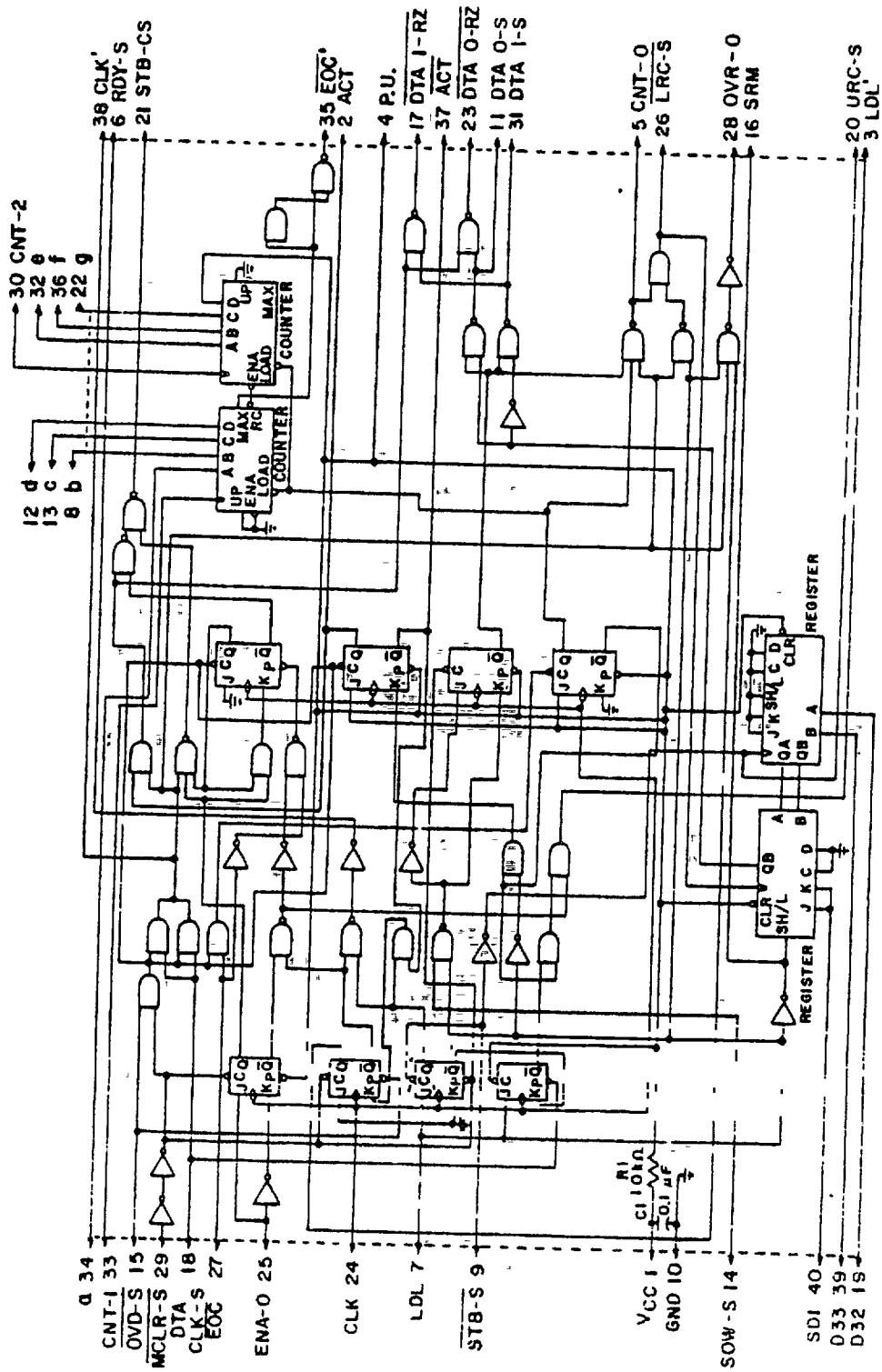



FIGURE 1. DQD module function diagram, standard serial output.



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TABLE I Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation | III | S | 10 | ∞ | 1.0 | ∞ | 10 | ∞ | 1.0 | ∞ | MΩ |
| B1-B16 | $V_{OUT}$ (low loaded) | IV | S | 0.00 | 0.80 | 0.00 | 0 80 | 0.00 | 0.80 | 0 00 | 0.80 | V |
| C1-C13 | Input breakdown current | V | S | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0 0 | 1.0 | mA |
| C14-C18 | Input breakdown current | V | S | 0.0 | 2 0 | 0.0 | 2 0 | 0.0 | 2.0 | 0.0 | 2.0 | mA |
| C19 | Input breakdown current | V | S | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | mA |
| C20 | Input breakdown current | V | S | 0.0 | 9 0 | 0 0 | 9.0 | 0.0 | 9.0 | 0.0 | 9.0 | mA |
| C21 | Input breakdown current | V | S | 0.0 | 8.0 | 0.0 | 8.0 | 0.0 | 8.0 | 0.0 | 8.0 | mA |
| D1-D15 | Output short circuit current | VI | X | -20 | -60 | -20 | -60 | -20 | -60 | -20 | -60 | mA |
| D16 | Output short circuit current | VI | X | -40 | -120 | -40 | -120 | -40 | -120 | -40 | -120 | mA |
| E1-E146 | Functional | VII | X | Test vectors must produce output vectors as specified in table VII | | | | | | | | |
| F1-F10, F12-F21 | Input clamp diode voltage | VIII | X | -0.4 | -1.5 | -0.4 | -1.5 | -0 4 | -1 5 | -0.4 | -1.5 | V |
| F11 | Input clamp diode voltage | VIII | X | -0 1 | -1.5 | -0.1 | -1.5 | -0.1 | -1.5 | -0 1 | -1 5 | V |
| G1-G16 | Static output voltage, low level | IX | X | 0.01 | 0.40 | 0 01 | 0.40 | 0.01 | 0 40 | 0 01 | 0.40 | V |
| H1-H16 | Static output voltage, high level | X | X | 2.4 | 4 5 | 2.4 | 4 5 | 2.4 | 4 5 | 2.4 | 4.5 | V |
| J1, J2 | Input current, high level | XI | X | 0 | 20 | 0 | 20 | 0 | 20 | 0 | 20 | µA |
| J3-J14 | Input current, high level | XI | X | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | µA |

See footnotes at end of table



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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| J15-J17 | Input current, high level | XI | X | 0 | 80 | 0 | 80 | 0 | 80 | 0 | 80 | µA |
| J18,J19 | Input current, high level | XI | X | 0 | 120 | 0 | 120 | 0 | 120 | 0 | 120 | µA |
| J20 | Input current, high level | XI | X | 0 | 360 | 0 | 360 | 0 | 360 | 0 | 360 | µA |
| J21 | Input current, high level | XI | X | -18,000 | 640 | -18,000 | 640 | -18,000 | 640 | -18,000 | 640 | µA |
| K1, K2 | Input current, low level | XII | X | -0.100 | -0.389 | -0.100 | -0.389 | -0.100 | -0.389 | -0.100 | -0.389 | mA |
| K3 | Input current, low level | XII | X | -0.100 | -0.778 | -0.100 | -0.778 | -0.100 | -0.778 | -0.100 | -0.778 | mA |
| K4-K14 | Input current, low level | XII | X | -0.10 | -1.73 | -0.10 | -1.73 | -0.10 | -1.73 | -0.10 | -1.73 | mA |
| K15,K16 | Input current, low level | XII | X | -0.10 | -4.80 | -0.10 | -4.80 | -0.10 | -4.80 | -0.10 | -4.80 | mA |
| K17-K19 | Input current, low level | XII | X | -0.10 | -3.46 | -0.10 | -3.46 | -0.10 | -3.46 | -0.10 | -3.46 | mA |
| K20 | Input current, low level | XII | X | -0.10 | -15.6 | -0.10 | -15.6 | -0.10 | -15.6 | -0.10 | -15.6 | mA |
| K21 | Input current, low level | XII | X | -0.10 | -25.6 | -0.10 | -25.6 | -0.10 | -25.6 | -0.10 | -25.6 | mA |
| L | Power supply current | XIII | X | 130 | 520 | 130 | 520 | 130 | 520 | 130 | 520 | mA |
| M1-M16 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | XIV | S* | 4 | 30 | 4 | 30 | 4 | 35 | 4 | 35 | ns |
| M17-M32 | Transition time ($t_{TLH}$), $C_L$ = 150 pF | XIV | Q | 4 | 50 | 4 | 50 | 4 | 55 | 4 | 55 | ns |

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| N1-N16 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XV | S | 4 | 13 | 4 | 13 | 4 | 15 | 4 | 15 | ns |
| N17-N32 | Transition time ($t_{THL}$), $C_L$ = 15 pF | XV | Q | 4 | 20 | 4 | 20 | 4 | 23 | 4 | 23 | ns |
| P1 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVI | S | 20 | 122 | 20 | 122 | 20 | 127 | 20 | 127 | ns |
| P2, P4 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVI | S | 12 | 64 | 12 | 64 | 12 | 69 | 12 | 69 | ns |
| P3 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVI | S | 16 | 95 | 16 | 95 | 16 | 100 | 16 | 100 | ns |
| P5 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVI | S | 12 | 72 | 12 | 72 | 12 | 77 | 12 | 77 | ns |
| P6 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVI | S | 24 | 131 | 24 | 131 | 24 | 140 | 24 | 140 | ns |
| P7 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XVI | Q | 20 | 127 | 20 | 127 | 20 | 135 | 20 | 135 | ns |

See footnotes at end of table

MIL-M-38510/121A(EC)

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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| P8, P10 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XVI | Q | 12 | 69 | 12 | 69 | 12 | 73 | 12 | 73 | ns |
| P9 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XVI | Q | 16 | 100 | 16 | 100 | 16 | 105 | 16 | 105 | ns |
| P11 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XVI | Q | 12 | 77 | 12 | 77 | 12 | 81 | 12 | 81 | ns |
| P12 | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | XVI | Q | 24 | 136 | 24 | 136 | 24 | 145 | 24 | 145 | ns |
| Q1 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XVII | S | 20 | 108 | 20 | 108 | 20 | 115 | 20 | 115 | ns |
| Q2, Q4 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XVII | S | 12 | 72 | 12 | 72 | 12 | 75 | 12 | 75 | ns |
| Q3 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XVII | S | 16 | 85 | 16 | 85 | 16 | 90 | 16 | 90 | ns |
| Q5 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XVII | S | 12 | 64 | 12 | 64 | 12 | 68 | 12 | 68 | ns |
| Q6 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XVII | S | 24 | 136 | 24 | 136 | 24 | 140 | 24 | 140 | ns |
| Q7 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XVII | Q | 20 | 115 | 20 | 115 | 20 | 120 | 20 | 120 | ns |

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

| Test condition | Parameter | Ref test table | Test type 1/-3/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Q8, Q10 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XVII | Q | 12 | 79 | 12 | 79 | 12 | 83 | 12 | 83 | ns |
| Q9 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XVII | Q | 16 | 94 | 16 | 94 | 16 | 100 | 16 | 100 | ns |
| Q11 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XVII | Q | 12 | 71 | 12 | 71 | 12 | 74 | 12 | 74 | ns |
| Q12 | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | XVII | Q | 24 | 143 | 24 | 143 | 24 | 150 | 24 | 150 | ns |
| R1 | Skew | XVIII | S | -6 | 100 | -6 | 100 | -6 | 100 | -6 | 100 | ns |
| R2 | Skew | XVIII | S | -5 | 5 | -5 | 5 | -5 | 5 | -5 | 5 | ns |
| S1 | Spike width | XVIII | S | 0 | 3 | 0 | 3 | 0 | 3 | 0 | 3 | ns |
| S2 | Pulse width | XVIII | S | 40 | 150 | 40 | 150 | 40 | 150 | 40 | 150 | ns |
| T1-T15 | Output crosstalk, high level | XIX | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| U1-U16 | Output crosstalk, low level | XX | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| V | Power supply transient amplitude | XXI | Q | 0 | 600 | 0 | 600 | 0 | 600 | 0 | 600 | mV |
| W1-W13 | Input capacitance | XXII | Q | 2 | 15 | 2 | 15 | 2 | 15 | 2 | 15 | pF |
| W14-W19 | Input capacitance | XXII | Q | 2 | 25 | 2 | 25 | 2 | 25 | 2 | 25 | pF |
| W20 | Input capacitance | XXII | Q | 2 | 120 | 2 | 120 | 2 | 120 | 2 | 120 | pF |
| W21 | Input capacitance | XXII | Q | 2 | 95 | 2 | 95 | 2 | 95 | 2 | 95 | pF |
| X | Power supply filter capacitance | XXIII | S | 0.07 | 0.13 | 0.07 | 0.13 | 0.07 | 0.13 | 0.07 | 0.13 | µF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests D through V and X shall be performed in the order specified in 4 4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross-reference of classification. Part number M28787/212-1 supersedes part number M28787/212. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

| | | |
|---|---|---|
| Maximum power dissipation | | 2,860 mW (outputs not loaded) |
| Supply voltage | | 5.0 Vdc ±10 percent |
| Input voltage | | |
| Low level (all except SDI, D32, and D33) | | -0.5 V minimum, 0.8 V maximum |
| Low level (SDI, D32, and D33) | | -0.5 V minimum, 0.7 V maximum |
| High level | | 2.0 V minimum, $V_{CC}$ maximum |
| Input current | | |
| D32 and D33 | | |
| Low level | ($V_{IL}$ = 0.0 V) | -0.389 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -0.36 mA maximum |
| High level | ($V_{IH}$ = 2.7 V) | 20 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 0.1 mA maximum |
| SDI | | |
| Low level | ($V_{IL}$ = 0.0 V) | -0.778 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -0.72 mA maximum |
| High level | ($V_{IH}$ = 2.7 V) | 40 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 0.2 mA maximum |
| a-g, SOW-S, $\overline{\text{MCLR-S}}$, CNT-1, and CNT-2 | | |
| Low level | ($V_{IL}$ = 0.0 V) | -1.73 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -1.6 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 40 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 1.0 mA maximum |
| LOL | | |
| Low level | ($V_{IL}$ = 0.0 V) | -15.6 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -14.4 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 360 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 9.0 mA maximum |



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| | | |
|---|---|---|
| ENA-O, DTA CLK-S, and $\overline{EOC}$ | | |
| Low level | ($V_{IL}$ = 0.0 V) | -3.46 mA maximum |
| | ($V_{IL}$ = 0.4 V) | -3.2 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 80 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 2.0 mA maximum |
| $\overline{STB-S}$ and $\overline{OVD-S}$ | | |
| Low level | ($V_{IL}$ = 0.4 V) | -4.80 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 120 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 2.0 mA maximum |
| CLK | | |
| Low level | ($V_{IL}$ = 0.4 V) | -25.6 mA maximum |
| High level | ($V_{IH}$ = 2.4 V) | 640 µA maximum |
| | ($V_{IH}$ = 5.5 V) | 8.0 mA maximum |
| Output drive current maximum capability | | |
| CLK' | | |
| Low level | ($V_{OL}$ = 0.4 V) | 24 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -1,400 µA maximum |
| OVR-O, $\overline{DTA\ O-RZ}$, $\overline{DTA\ 1-RZ}$, STB-CS, $\overline{EOC}$', and RDY-S | | |
| Low level | ($V_{OL}$ = 0.4 V) | 16 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -400 µA maximum |
| LDL' | | |
| Low level | ($V_{OL}$ = 0.4 V) | 16 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -800 µA maximum |
| $\overline{LRC-S}$ | | |
| Low level | ($V_{OL}$ = 0.4 V) | 15.6 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -780 µA maximum |
| URC-S and SRM | | |
| Low level | ($V_{OL}$ = 0.4 V) | 15.6 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -380 µA maximum |
| DTA 1-S, DTA O-S, ACT, $\overline{ACT}$, and CNT-O | | |
| Low level | ($V_{OL}$ = 0.4 V) | 14.4 mA maximum |
| High level | ($V_{OH}$ = 2.4 V) | -360 µA maximum |
| Output voltage | | |
| CLK' | | |
| Low level | ($I_{OL}$ = 24 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -1,400 µA) | 2.4 V minimum |
| OVR-O, $\overline{DTA\ O-RZ}$, $\overline{DTA\ 1-RZ}$, STB-CS, $\overline{EOC}$', and RDY-S | | |
| Low level | ($I_{OL}$ = 16 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -400 µA) | 2.4 V minimum |
| LDL' | | |
| Low level | ($I_{OL}$ = 16 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -800 µA) | 2.4 V minimum |
| $\overline{LRC-S}$ | | |
| Low level | ($I_{OL}$ = 15.6 mA) | 0.4 V maximum |
| High level | ($I_{OH}$ = -780 µA) | 2.4 V minimum |



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| | |
|---|---|
| URC-S and SRM | |
| Low level ($I_{OL}$ = 15.6 mA) | 0.4 V maximum |
| High level ($I_{OH}$ = -380 µA) | 2.4 V minimum |
| DTA 1-S, DTA 0-S, ACT, $\overline{ACT}$, and CNT-0 | |
| Low level ($I_{OL}$ = 14.4 mA) | 0.4 V maximum |
| High level ($I_{OH}$ = -360 µA) | 2.4 V minimum |
| Propagation delay time ($t_{PLH}$) | |
| DTA CLK-S to DTA 0-S | |
| $C_L$ = 50 pF | 20 ns minimum, 127 ns maximum |
| $C_L$ = 150 pF | 20 ns minimum, 135 ns maximum |
| DTA CLK-S to $\overline{DTA\ 1\text{-}RZ}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 69 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 73 ns maximum |
| DTA CLK-S to STB-CS | |
| $C_L$ = 50 pF | 16 ns minimum, 100 ns maximum |
| $C_L$ = 150 pF | 16 ns minimum, 105 ns maximum |
| DTA CLK-S to $\overline{DTA\ 0\text{-}RZ}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 69 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 73 ns maximum |
| DTA CLK-S to $\overline{LRC\text{-}S}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 77 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 81 ns maximum |
| DTA CLK-S to DTA 1-S | |
| $C_L$ = 50 pF | 24 ns minimum, 140 ns maximum |
| $C_L$ = 150 pF | 24 ns minimum, 145 ns maximum |
| Propagation delay time ($t_{PHL}$) | |
| DTA CLK-S to DTA 0-S | |
| $C_L$ = 50 pF | 20 ns minimum, 115 ns maximum |
| $C_L$ = 150 pF | 20 ns minimum, 120 ns maximum |
| DTA CLK-S to $\overline{DTA\ 1\text{-}RZ}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 75 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 83 ns maximum |
| DTA CLK-S to STB-CS | |
| $C_L$ = 50 pF | 16 ns minimum, 90 ns maximum |
| $C_L$ = 150 pF | 16 ns minimum, 100 ns maximum |
| DTA CLK-S to $\overline{DTA\ 0\text{-}RZ}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 75 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 83 ns maximum |
| DTA CLK-S to $\overline{LRC\text{-}S}$ | |
| $C_L$ = 50 pF | 12 ns minimum, 68 ns maximum |
| $C_L$ = 150 pF | 12 ns minimum, 74 ns maximum |
| DTA CLK-S to DTA 1-S | |
| $C_L$ = 50 pF | 24 ns minimum, 140 ns maximum |
| $C_L$ = 150 pF | 24 ns minimum, 150 ns maximum |



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6.5 Module function. The standard serial output element control (MIL-M-28787/212) and the standard serial input element control (MIL-M-28787/215) are universal control modules which can be used in conjunction with a parallel to serial converter (MIL-M-28787/213) and a serial to parallel converter (MIL-M-28787/214) to make up the Standard Serial Output Element (SSOE) and the Standard Serial Input Element (SSIE). The SSOE and SSIE can be used in conjunction with line drivers and receivers to make up a serial data transmission system. The system can operate at data rates up to 1 megabit per second asynchronously or 2 megabits per second synchronously.

6.5.1 Transmission types. There are four serial transmission types possible when the SSOE and SSIE modules are used. These types include asynchronous or synchronous transmission each of which can be either controlled or uncontrolled. The asynchronous transmission differs from the synchronous type in that the former accomplishes message transfer without the necessity of a synchronizing clock line from the transmitter to the receiver. The control of either transmission type is accomplished by the use of discrete control lines connected between the appropriate subsystems.

6.5.2 Input to output relationship. The standard serial output control module has the input to output relationships shown on figures 2, 3, and 4. The length of the data word is programmed according to table II.

6.5.3 Input and output definitions. The interface to the SSOE consists of several types of general interfaces. These include user controls, clocks and strobes, data paths, test point jumpers, and serial line interfaces.

6.5.3.1 User controls. The user controls consist of $\overline{\text{MCLR-S}}$, $\overline{\text{OVD-S}}$, SOW-S, RDY-S, ACT, $\overline{\text{ACT}}$, SRM, and WORD LENGTH.

$\overline{\text{MCLR-S}}$: MASTER CLEAR is an active low input pulse that asynchronously clears the SSOE. If word transmission is in progress, it will be halted and the SSOE will remain idle until the next external clock strobe ($\overline{\text{STB-S}}$) signal.

$\overline{\text{OVD-S}}$: OVERRIDE is an active low input pulse that asynchronously clears the SSOE transmission. The $\overline{\text{OVD-S}}$ is usually used to stop transmission, clear both the sender and receiver, then force a word into the receiver register when used in conjunction with clock strobe ($\overline{\text{STB-S}}$). The $\overline{\text{OVD-S}}$ is normally used in conjunction with an input data strobe ($\overline{\text{STB-S}}$) to transmit a forced OVERRIDE word across the serial interface. $\overline{\text{OVD-S}}$ and $\overline{\text{STB-S}}$ are independent of the RDY-S signal or internal SSOE timing. There is a requirement the $\overline{\text{OVD-S}}$ and $\overline{\text{STB-S}}$ overlap one another in a time sequence for proper operation.

SOW-S. START OF WORD SELECT is the level which selects a character to be sent in the serial data stream which indicates a start of word. If SOW-S is high then the SOW bit is selected and appears in the serial data as the first bit of each word.

RDY-S. READY is an active high output level which indicates to the user that the SSOE holding register is empty. New data can be loaded into the holding register any time RDY-S is active. Loading the register is accomplished with the use of $\overline{\text{STB-S}}$.

ACT and $\overline{\text{ACT}}$: OUTPUT ACTIVE is an active high output level ACT is a high level while data is transmitted from the SSOE. As long as the enable (ENA-0) is high, ACT remains high for all data bits including "end of word". If "end of word" is delayed as noted in the ENA-0 description, ACT will go low one bit time after the last data bit and remain low until a new word is started. $\overline{\text{ACT}}$ is the negated function of ACT



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SRM. SHIFT REGISTER MODE control is an output signal level used to control the parallel and serial modes of the data registers. If the EQG register modules are used, a high level puts the register in the serial or shift mode, a low level puts it in a load or parallel mode.

WORD LENGTH: These are seven inputs (a-g) which usually are pin programmed to identify the number of bits which are to be contained in each serial word that will be transferred across the serial interface. In the controlled modes, the SSOE will be programmed to the number of bits/word (see table II). In the broadcast mode (uncontrolled asynchronous transmission) the SSOE will be programmed to one count less than the number of bits/word (see table II). The range of serial word lengths range from 1 to 128 bits in length. The usual length in the fire control system is 32 bits and will be programmed a=0, b=0, c=0, d=0, e=0, f=1, and g=1 for asynchronous transmission.

6.5 3.2 Clocks and strobes. The clock and strobes interface to the SSOE consists of DTA CLK-S, $\overline{\text{STB-S}}$, URC-S and $\overline{\text{LRC-S}}$.

DTA CLK-S DATA CLOCK is an external clock input used to set the transmission data bit rate and pulse width on the DTA 1-S and DTA 0-S lines. This clock input is asynchronous with respect to other SSOE inputs.

$\overline{\text{STB-S}}$. DATA STROBE is an input active low strobe which causes loading of data into the SSOE holding register by generating URC-S and will initiate all data transmission from the SSOE. Except for the data override option, $\overline{\text{STB-S}}$ must not be applied to the SSOE unless the ready signal (RDY-S) is at a high level.

URC-S: UPPER RANK CLOCK is an active high clock output generated by $\overline{\text{STB-S}}$ and is used to load new data into the external holding register and the SSOE internal holding register (2 bits).

$\overline{\text{LRC-S}}$: LOWER RANK CLOCK is an active low clock output which is used to load the SSOE shift register and to shift that data out onto the serial interface. Shifting occurs on the positive (trailing) edge. $\overline{\text{LRC-S}}$ is generated by DTA CLK-S when shifting data. This determines bit frequency of serial transmission.

6.5.3 3 Data paths. The data paths excluding the serial interface include D33, D32, and SDI.

D33. DATA BIT 33 is the MSB (most significant bit) of the word entering the SSOE module in parallel from the source. This bit is transferred across the serial interface. In general, this would be the COMMAND bit.

D32 DATA BIT 32 is the next bit in succession from the MSB to the LSB (least significant bit) entering the SSOE module in parallel from the source. This will be the second bit within a word to be transferred across the serial interface.

SDI SERIAL DATA INPUT is an input line over which all serial data is transferred from the external register module into the SSOE module. This data eventually is sent out on the serial interface SDI is connected, in general, to the SDO (SERIAL DATA OUTPUT) pin of the EQG register module



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6.5.3.4 Test point jumpers. The test point jumpers consist of CLK', CLK, LDL', LDL, $\overline{EOC}$', $\overline{EOC}$, CNT-0, CNT-1, CNT-2, and P.U. All names with (') sign are outputs and the names without the (') sign are inputs except CNT-0 and P.U. which are outputs. Test point jumper inputs and outputs are for test purposes only.

P.U.: PULL UP is a static high logic level output signal which is available to the user for other modules if the appropriate conditions exist. There is a drive capability for 30 unit loads when the user is in the high state. A unit load = 40 µA.

6.5.3.5 Serial line interfaces. The serial line interfaces include ENA-0, OVR-0, $\overline{DTA\ 1\text{-}RZ}$, $\overline{DTA\ 0\text{-}RZ}$, DTA 1-S, DTA 0-S, and STB-CS.

ENA-0 ENABLE is an active high input level which, in general, is controlled by the receiver element for the purpose of word frequency control. Where the ENABLE is active high, the SSOE is enabled to transmit data. If the ENABLE is not active, the SSOE cannot transmit data unless an OVR-0 is sent. The following is a more detailed listing of how ENA-0 functions:

a. ENA-0 at high level is a necessary condition for SSOE normal word transmission. An OVERRIDE word can be transmitted independent of ENA-0 state.

b. If ENA-0 is changed to low level during a word transmission in the synchronous mode the current word will be completed except for the last bit. This last bit will be transmitted when ENA-0 becomes active again. The last bit is considered to be an "end of word" (EOW) bit for synchronous transmission. In asynchronous mode, the last data bit (LSB) is always transmitted independent of ENA-0.

OVR-0 OVERRIDE is an output active high pulse which is used as a forcing function to clear and force a word into the receiver. A return-to-zero (RZ) pulse will appear on the OVR-0 line within two clock periods (DTA CLK-S) after it has been initiated by $\overline{OVD\text{-}S}$. OVR-0 will be coincident with the active high portion of DTA CLK-S. In general, OVR-0 is initiated and sent coincident with $\overline{STB\text{-}S}$. If $\overline{STB\text{-}S}$ is not present, the data word following the OVR-0 pulse will be data contained in the SSOE shift register from previous transmission.

$\overline{DTA\ 1\text{-}RZ}$ DATA 1 with RETURN TO ZERO coding is an active low output signal which in general is inputted into differential line drivers to transmit serial data asynchronously across the serial interface. $\overline{DTA\ 1\text{-}RZ}$ represents the negated TRUE data from the SSOE data registers and is active only during start of word bit times and when high levels are transmitted across the interface.

$\overline{DTA\ 0\text{-}RZ}$ DATA 0 with RETURN TO ZERO coding is an active low output signal which in general is inputted into differential line drivers to transmit serial data asynchronously across the serial interface. $\overline{DTA\ 0\text{-}RZ}$ represents the negated FALSE data from the SSOE data registers and is active only during start of word bit times and when low levels are sent across the interface.

DTA 1-S DATA 1 with NON RETURN TO ZERO (NRZ) coding is an active high output signal which in general is inputted into differential line drivers to transmit serial data synchronously across the serial interface. DTA 1-S represents the TRUE data from the SSOE data registers and is active only during start of word bit times and where high levels are sent across the interface.



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DTA 0-S: DATA 0 with NON RETURN TO ZERO (NRZ) coding is an active high output signal which in general is not used. If used it is inputted into differential line drivers to transmit serial data at the user's discretion. DTA 0-S represents the FALSE data from the SSOE data registers and is active only during start of word bit times and when low levels are sent across the interface.

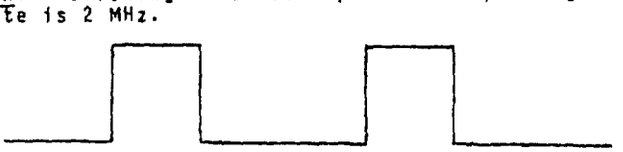
STB-CS: STROBE (CONTROLLED SERIAL) is an active high output pulse train which in general is used in the synchronous transmission mode to clock (synchronize) the receiver electronics and the data across the serial interface. STB-CS is coincident and generated by the DTA CLK-S input clock. STB-CS is active only during message transmission.

6.5.4 Pin nomenclature. Module pin assignments and the function description are as follows.

| Function | Pin No. | Description |
|---|---|---|
| ENA-O | 25 | Enable (input) |
| $\overline{\text{MCLR-S}}$ | 29 | Master clear (input) |
| $\overline{\text{STB-S}}$ | 9 | Data strobe (input) |
| $\overline{\text{OVD-S}}$ | 15 | Override (input) |
| DTA CLK-S | 18 | Data clock (input) |
| CLK' | 38 | Clock (output test point jumper) |
| CLK | 24 | Clock (input test point jumper) |
| ACT | 2 | Output active (output) |
| $\overline{\text{ACT}}$ | 37 | Output active complement (output) |
| LDL' | 3 | (Output test point jumper) |
| LDL | 7 | (Input test point jumper) |
| URC-S | 20 | Upper rank clock (output) |
| SDI | 40 | Serial data input (input) |
| SOW-S | 14 | Start of word select (input) |
| RDY-S | 6 | Ready (output) |
| $\overline{\text{EOC}}$' | 35 | End of count (output test point jumper) |
| $\overline{\text{EOC}}$ | 27 | End of count (input test point jumper) |
| D32,D33 | 19,39 | Data bits (inputs) |
| SRM | 16 | Shift register mode control (output) |
| DTA 1-S | 31 | Data 1 (NRZ) (output) |
| DTA 0-S | 11 | Data 0 (NRZ) (output) |
| $\overline{\text{DTA 1-RZ}}$ | 17 | Data 1 (RZ) (output) |
| $\overline{\text{DTA 0-RZ}}$ | 23 | Data 0 (RZ) (output) |
| STB-CS | 21 | Strobe (controlled serial) (output) |
| P.U. | 4 | Pull up (output test point jumper-internal) |
| $\overline{\text{OVR-O}}$ | 28 | Override (output) |
| $\overline{\text{LRC-S}}$ | 26 | Lower rank clock (output) |
| a,b,c,d,e,f,g | 34,8,13,12, 32,36,22 | Word length program lines (inputs) |
| CNT-0 | 5 | Bit counter (output test point jumper) |
| CNT-1 | 33 | Bit counter (input test point jumper) |
| CNT-2 | 30 | Bit counter (input test point jumper) |
| $V_{CC}$ | 1 | +5 V power |
| GND | 10 | +5 V return |

6.5.5 Signal input description. Following is a description of input signal parameters. The maximum data rate is 2 MHz.

DTA CLK-S



Maximum rate = 2 MHz.
Clock high (minimum) = 220 ns.
Clock low (minimum) = 280 ns.

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Pulse width (minimum) = 50 ns

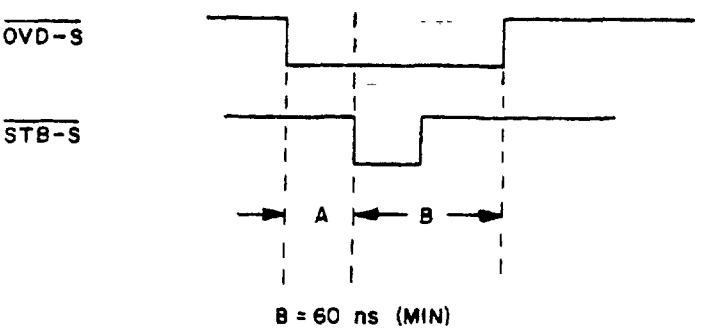



Pulse width (minimum) = 50 ns.
Pulse width (maximum) = DTA CLK-S period minus 20 ns




Pulse width (minimum) = 60 ns + A (see below)

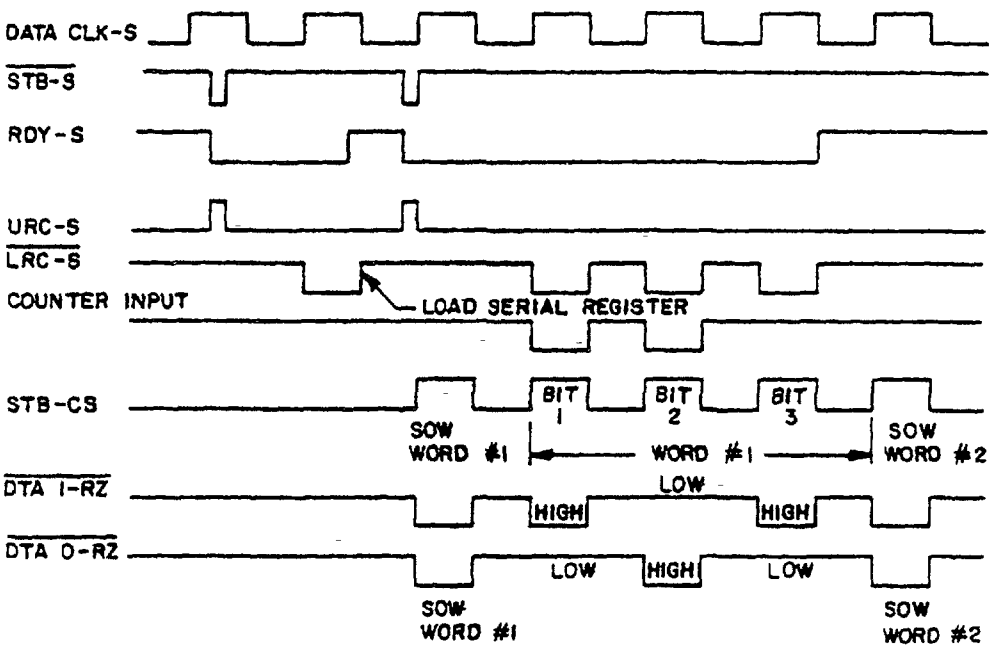
$\overline{\text{OVD-S}}$ must be accompanied by a $\overline{\text{STB-S}}$ pulse with the following relationship




6.5.6 Parallel outputs. Paralleling of outputs is not permissible.



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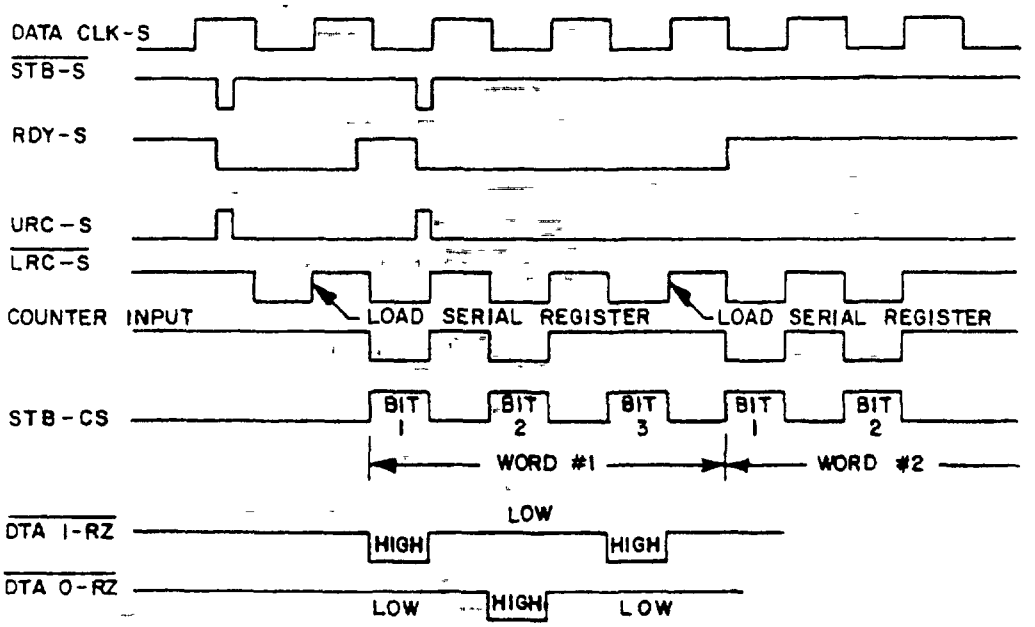



NOTES:
1. Word length = 3 bits.
   Counter set to "3". Transmission will be 3 bits plus SOW bit (from table II).
2. Parallel input data:
   D33 = high level.
   D32 = low level.
   D31 = high level.
3. STB-CS is shown for reference only.

FIGURE 2. Broadcast (mode timing)



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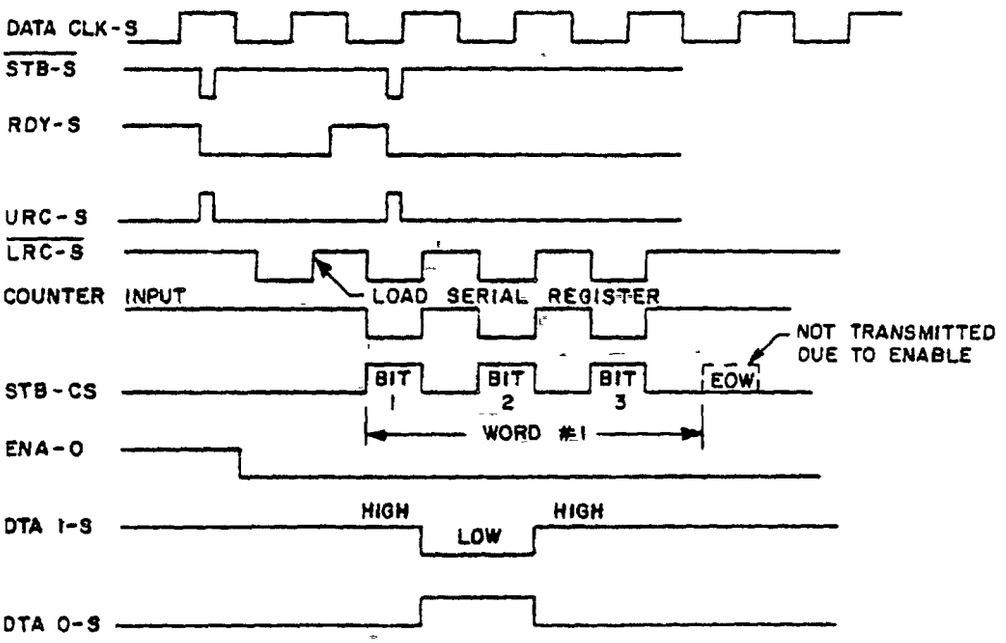



NOTES
1. Word length = 3 bits.
   Counter set to "3". Transmission will be 3 bits (from table II). No SOW.
2. Parallel input data
   D33 = high level.
   D32 = low level.
   D31 = high level.
3. STB-CS is shown for reference only.

FIGURE 3 Broadcast



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NOTE: Word length = 3 bits.
Counter set to "3" (from table II).

FIGURE 4. Controlled serial mode timing.



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TABLE II. Counter pin programming (SSOE).

| Mode - Broadcast | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | H | H | H | H | H | H | 0 |
| L | H | H | H | H | H | H | 2 |
| H | L | H | H | H | H | H | 3 |
| L | L | H | H | H | H | H | 4 |
| H | H | L | H | H | H | H | 5 |
| L | H | L | H | H | H | H | 6 |
| H | L | L | H | H | H | H | 7 |
| L | L | L | H | H | H | H | 8 |
| H | H | H | L | H | H | H | 9 |
| L | H | H | L | H | H | H | 10 |
| H | L | H | L | H | H | H | 11 |
| L | L | H | L | H | H | H | 12 |
| H | H | L | L | H | H | H | 13 |
| L | H | L | L | H | H | H | 14 |
| H | L | L | L | H | H | H | 15 |
| L | L | L | L | H | H | H | 16 |
| H | H | H | H | L | H | H | 17 |
| L | H | H | H | L | H | H | 18 |
| H | L | H | H | L | H | H | 19 |
| L | L | H | H | L | H | H | 20 |
| H | H | L | H | L | H | H | 21 |
| L | H | L | H | L | H | H | 22 |
| H | L | L | H | L | H | H | 23 |
| L | L | L | H | L | H | H | 24 |
| H | H | H | L | L | H | H | 25 |
| L | H | H | L | L | H | H | 26 |
| H | L | H | L | L | H | H | 27 |
| L | L | H | L | L | H | H | 28 |
| H | H | L | L | L | H | H | 29 |
| L | H | L | L | L | H | H | 30 |
| H | L | L | L | L | H | H | 31 |
| L | L | L | L | L | H | H | 32 |
| H | H | H | H | H | L | H | 33 |
| L | H | H | H | H | L | H | 34 |
| H | L | H | H | H | L | H | 35 |
| L | L | H | H | H | L | H | 36 |
| H | H | L | H | H | L | H | 37 |
| L | H | L | H | H | L | H | 38 |
| H | L | L | H | H | L | H | 39 |
| L | L | L | H | H | L | H | 40 |
| H | H | H | L | H | L | H | 41 |
| L | H | H | L | H | L | H | 42 |
| H | L | H | L | H | L | H | 43 |
| L | L | H | L | H | L | H | 44 |
| H | H | L | L | H | L | H | 45 |
| L | H | L | L | H | L | H | 46 |
| H | L | L | L | H | L | H | 47 |
| L | L | L | L | H | L | H | 48 |
| H | H | H | H | L | L | H | 49 |
| L | H | H | H | L | L | H | 50 |

See footnotes at end of table



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TABLE II. Counter pin programming (SSOL) - Continued.

| Mode - Broadcast | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | L | H | H | L | L | H | 51 |
| L | L | H | H | L | L | H | 52 |
| H | H | L | H | L | L | H | 53 |
| L | H | L | H | L | L | H | 54 |
| H | L | L | H | L | L | H | 55 |
| L | L | L | H | L | L | H | 56 |
| H | H | H | L | L | L | H | 57 |
| L | H | H | L | L | L | H | 58 |
| H | L | H | L | L | L | H | 59 |
| L | L | H | L | L | L | H | 60 |
| H | H | L | L | L | L | H | 61 |
| L | H | L | L | L | L | H | 62 |
| H | L | L | L | L | L | H | 63 |
| L | L | L | L | L | L | H | 64 |
| H | H | H | H | H | H | L | 65 |
| L | H | H | H | H | H | L | 66 |
| H | L | H | H | H | H | L | 67 |
| L | L | H | H | H | H | L | 68 |
| H | H | L | H | H | H | L | 69 |
| L | H | L | H | H | H | L | 70 |
| H | L | L | H | H | H | L | 71 |
| L | L | L | H | H | H | L | 72 |
| H | H | H | L | H | H | L | 73 |
| L | H | H | L | H | H | L | 74 |
| H | L | H | L | H | H | L | 75 |
| L | L | H | L | H | H | L | 76 |
| H | H | L | L | H | H | L | 77 |
| L | H | L | L | H | H | L | 78 |
| H | L | L | L | H | H | L | 79 |
| L | L | L | L | H | H | L | 80 |
| H | H | H | H | L | H | L | 81 |
| L | H | H | H | L | H | L | 82 |
| H | L | H | H | L | H | L | 83 |
| L | L | H | H | L | H | L | 84 |
| H | H | L | H | L | H | L | 85 |
| L | H | L | H | L | H | L | 86 |
| H | L | L | H | L | H | L | 87 |
| L | L | L | H | L | H | L | 88 |
| H | H | H | L | L | H | L | 89 |
| L | H | H | L | L | H | L | 90 |
| H | L | H | L | L | H | L | 91 |
| L | L | H | L | L | H | L | 92 |
| H | H | L | L | L | H | L | 93 |
| L | H | L | L | L | H | L | 94 |
| H | L | L | L | L | H | L | 95 |
| L | L | L | L | L | H | L | 96 |
| H | H | H | H | H | L | L | 97 |
| L | H | H | H | H | L | L | 98 |
| H | L | H | H | H | L | L | 99 |
| L | L | H | H | H | L | L | 100 |

See footnotes at end of table.



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TABLE II. Counter pin programming (SSOE) - Continued.

| Mode - Broadcast | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | H | L | H | H | L | L | 101 |
| L | H | L | H | H | L | L | 102 |
| H | L | L | H | H | L | L | 103 |
| L | L | L | H | H | L | L | 104 |
| H | H | H | L | H | L | L | 105 |
| L | H | H | L | H | L | L | 106 |
| H | L | H | L | H | L | L | 107 |
| L | L | H | L | H | L | L | 108 |
| H | H | L | L | H | L | L | 109 |
| L | H | L | L | H | L | L | 110 |
| H | L | L | L | H | L | L | 111 |
| L | L | L | L | H | L | L | 112 |
| H | H | H | H | L | L | L | 113 |
| L | H | H | H | L | L | L | 114 |
| H | L | H | H | L | L | L | 115 |
| L | L | H | H | L | L | L | 116 |
| H | H | L | H | L | L | L | 117 |
| L | H | L | H | L | L | L | 118 |
| H | L | L | H | L | L | L | 119 |
| L | L | L | H | L | L | L | 120 |
| H | H | H | L | L | L | L | 121 |
| L | H | H | L | L | L | L | 122 |
| H | L | H | L | L | L | L | 123 |
| L | L | H | L | L | L | L | 124 |
| H | H | L | L | L | L | L | 125 |
| L | H | L | L | L | L | L | 126 |
| H | L | L | L | L | L | L | 127 |
| L | L | L | L | L | L | L | 128 |

See footnotes at end of table.



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TABLE II. Counter pin programming (SSOE) - Continued.

| Mode - Control 0 and Control 1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | H | H | H | H | H | H | 0 |
| L | H | H | H | H | H | H | 1 |
| H | L | H | H | H | H | H | 2 |
| L | L | H | H | H | H | H | 3 |
| H | H | L | H | H | H | H | 4 |
| L | H | L | H | H | H | H | 5 |
| H | L | L | H | H | H | H | 6 |
| L | L | L | H | H | H | H | 7 |
| H | H | H | L | H | H | H | 8 |
| L | H | H | L | H | H | H | 9 |
| H | L | H | L | H | H | H | 10 |
| L | L | H | L | H | H | H | 11 |
| H | H | L | L | H | H | H | 12 |
| L | H | L | L | H | H | H | 13 |
| H | L | L | L | H | H | H | 14 |
| L | L | L | L | H | H | H | 15 |
| H | H | H | H | L | H | H | 16 |
| L | H | H | H | L | H | H | 17 |
| H | L | H | H | L | H | H | 18 |
| L | L | H | H | L | H | H | 19 |
| H | H | L | H | L | H | H | 20 |
| L | H | L | H | L | H | H | 21 |
| H | L | L | H | L | H | H | 22 |
| L | L | L | H | L | H | H | 23 |
| H | H | H | L | L | H | H | 24 |
| L | H | H | L | L | H | H | 25 |
| H | L | H | L | L | H | H | 26 |
| L | L | H | L | L | H | H | 27 |
| H | H | L | L | L | H | H | 28 |
| L | H | L | L | L | H | H | 29 |
| H | L | L | L | L | H | H | 30 |
| L | L | L | L | L | H | H | 31 |
| H | H | H | H | H | L | H | 32 |
| L | H | H | H | H | L | H | 33 |
| H | L | H | H | H | L | H | 34 |
| L | L | H | H | H | L | H | 35 |
| H | H | L | H | H | L | H | 36 |
| L | H | L | H | H | L | H | 37 |
| H | L | L | H | H | L | H | 38 |
| L | L | L | H | H | L | H | 39 |
| H | H | H | L | H | L | H | 40 |
| L | H | H | L | H | L | H | 41 |
| H | L | H | L | H | L | H | 42 |
| L | L | H | L | H | L | H | 43 |
| H | H | L | L | H | L | H | 44 |
| L | H | L | L | H | L | H | 45 |
| H | L | L | L | H | L | H | 46 |
| L | L | L | L | H | L | H | 47 |
| H | H | H | H | L | L | H | 48 |
| L | H | H | H | L | L | H | 49 |

See footnotes at end of table.



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TABLE II. Counter pin programming (SSOE) - Continued.

| Mode - Control 0 and Control 1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | L | H | H | L | L | H | 50 |
| L | L | H | H | L | L | H | 51 |
| H | H | L | H | L | L | H | 52 |
| L | H | L | H | L | L | H | 53 |
| H | L | L | H | L | L | H | 54 |
| L | L | L | H | L | L | H | 55 |
| H | H | H | L | L | L | H | 56 |
| L | H | H | L | L | L | H | 57 |
| H | L | H | L | L | L | H | 58 |
| L | L | H | L | L | L | H | 59 |
| H | H | L | L | L | L | H | 60 |
| L | H | L | L | L | L | H | 61 |
| H | L | L | L | L | L | H | 62 |
| L | L | L | L | L | L | H | 63 |
| H | H | H | H | H | H | L | 64 |
| L | H | H | H | H | H | L | 65 |
| H | L | H | H | H | H | L | 66 |
| L | L | H | H | H | H | L | 67 |
| H | H | L | H | H | H | L | 68 |
| L | H | L | H | H | H | L | 69 |
| H | L | L | H | H | H | L | 70 |
| L | L | L | H | H | H | L | 71 |
| H | H | H | L | H | H | L | 72 |
| L | H | H | L | H | H | L | 73 |
| H | L | H | L | H | H | L | 74 |
| L | L | H | L | H | H | L | 75 |
| H | H | L | L | H | H | L | 76 |
| L | H | L | L | H | H | L | 77 |
| H | L | L | L | H | H | L | 78 |
| L | L | L | L | H | H | L | 79 |
| H | H | H | H | L | H | L | 80 |
| L | H | H | H | L | H | L | 81 |
| H | L | H | H | L | H | L | 82 |
| L | L | H | H | L | H | L | 83 |
| H | H | L | H | L | H | L | 84 |
| L | H | L | H | L | H | L | 85 |
| H | L | L | H | L | H | L | 86 |
| L | L | L | H | L | H | L | 87 |
| H | H | H | L | L | H | L | 88 |
| L | H | H | L | L | H | L | 89 |
| H | L | H | L | L | H | L | 90 |
| L | L | H | L | L | H | L | 91 |
| H | H | L | L | L | H | L | 92 |
| L | H | L | L | L | H | L | 93 |
| H | L | L | L | L | H | L | 94 |
| L | L | L | L | L | H | L | 95 |
| H | H | H | H | H | L | L | 96 |
| L | H | H | H | H | L | L | 97 |
| H | L | H | H | H | L | L | 98 |
| L | L | H | H | H | L | L | 99 |

See footnotes at end of table.



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TABLE II. Counter pin programming (SSOE) - Continued.

| Mode - Control 0 and Control 1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Module pins (see footnotes) | | | | | | | Word length (bits) |
| 34 | 8 | 13 | 12 | 32 | 36 | 22 | |
| H | H | L | H | H | L | L | 100 |
| L | H | L | H | H | L | L | 101 |
| H | L | L | H | H | L | L | 102 |
| L | L | L | H | H | L | L | 103 |
| H | H | H | L | H | L | L | 104 |
| L | H | H | L | H | L | L | 105 |
| H | L | H | L | H | L | L | 106 |
| L | L | H | L | H | L | L | 107 |
| H | H | L | L | H | L | L | 108 |
| L | H | L | L | H | L | L | 109 |
| H | L | L | L | H | L | L | 110 |
| L | L | L | L | H | L | L | 111 |
| H | H | H | H | L | L | L | 112 |
| L | H | H | H | L | L | L | 113 |
| H | L | H | H | L | L | L | 114 |
| L | L | H | H | L | L | L | 115 |
| H | H | L | H | L | L | L | 116 |
| L | H | L | H | L | L | L | 117 |
| H | L | L | H | L | L | L | 118 |
| L | L | L | H | L | L | L | 119 |
| H | H | H | L | L | L | L | 120 |
| L | H | H | L | L | L | L | 121 |
| H | L | H | L | L | L | L | 122 |
| L | L | H | L | L | L | L | 123 |
| H | H | L | L | L | L | L | 124 |
| L | H | L | L | L | L | L | 125 |
| H | L | L | L | L | L | L | 126 |
| L | L | L | L | L | L | L | 127 |

1/ L = Module pin grounded.
2/ H = Module pin tied to pullup resistor.



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6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

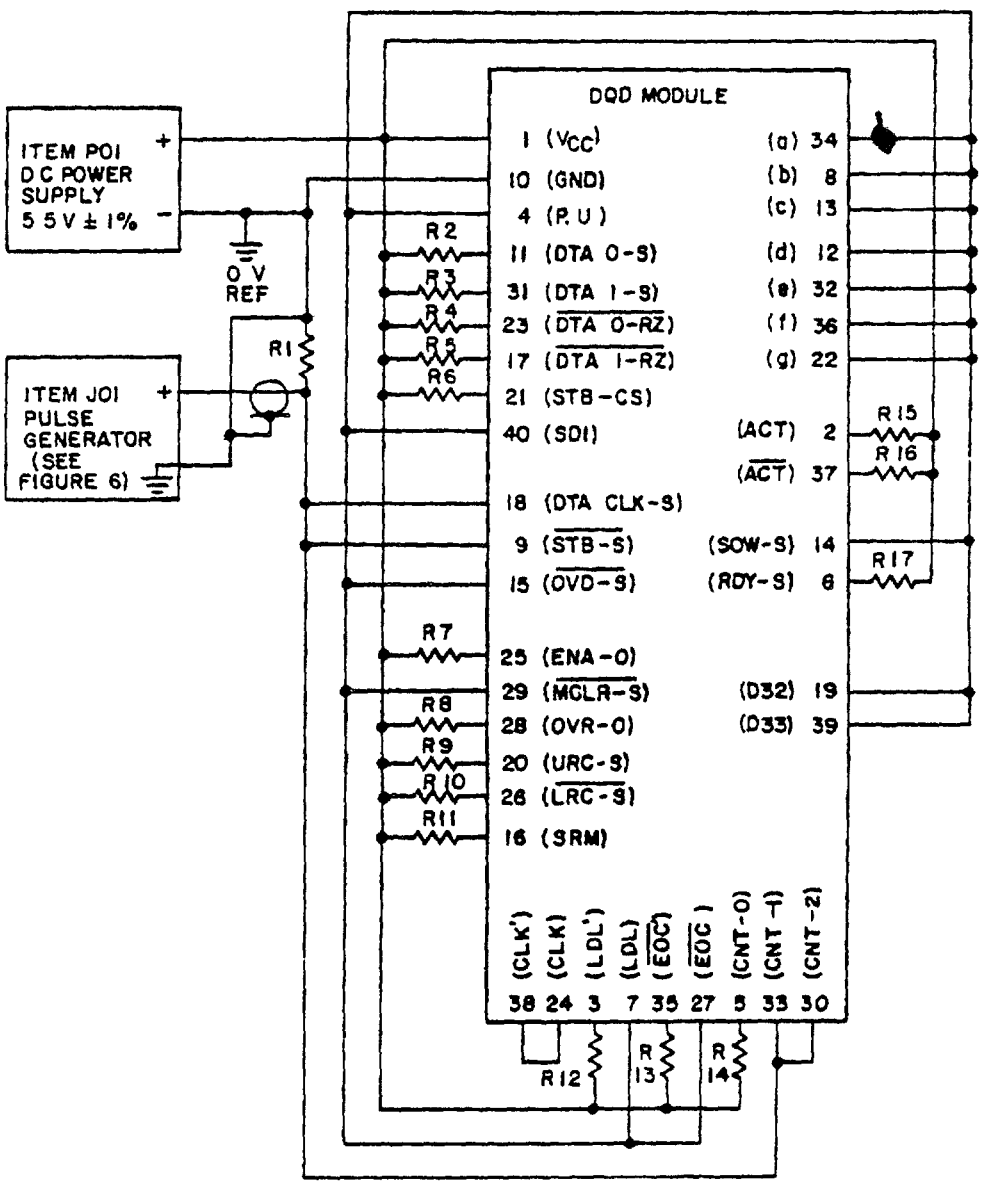
Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N348)



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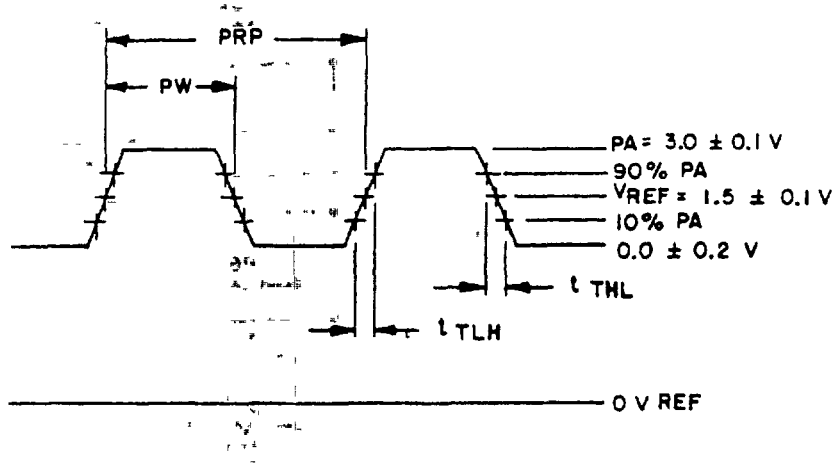



NOTES
1. R1 = 51 Ω ±1%, 1/4 W
2. R2 = R3 = R14-R16 = 357Ω ±1%, 1/4 W
3. R4-R8 = R12 = R13 = R17 = 316Ω ±1% 1/4 W
4. R9-R11 = 332Ω ±1%, 1/4 W

FIGURE 5. Operating temperature and life test circuit



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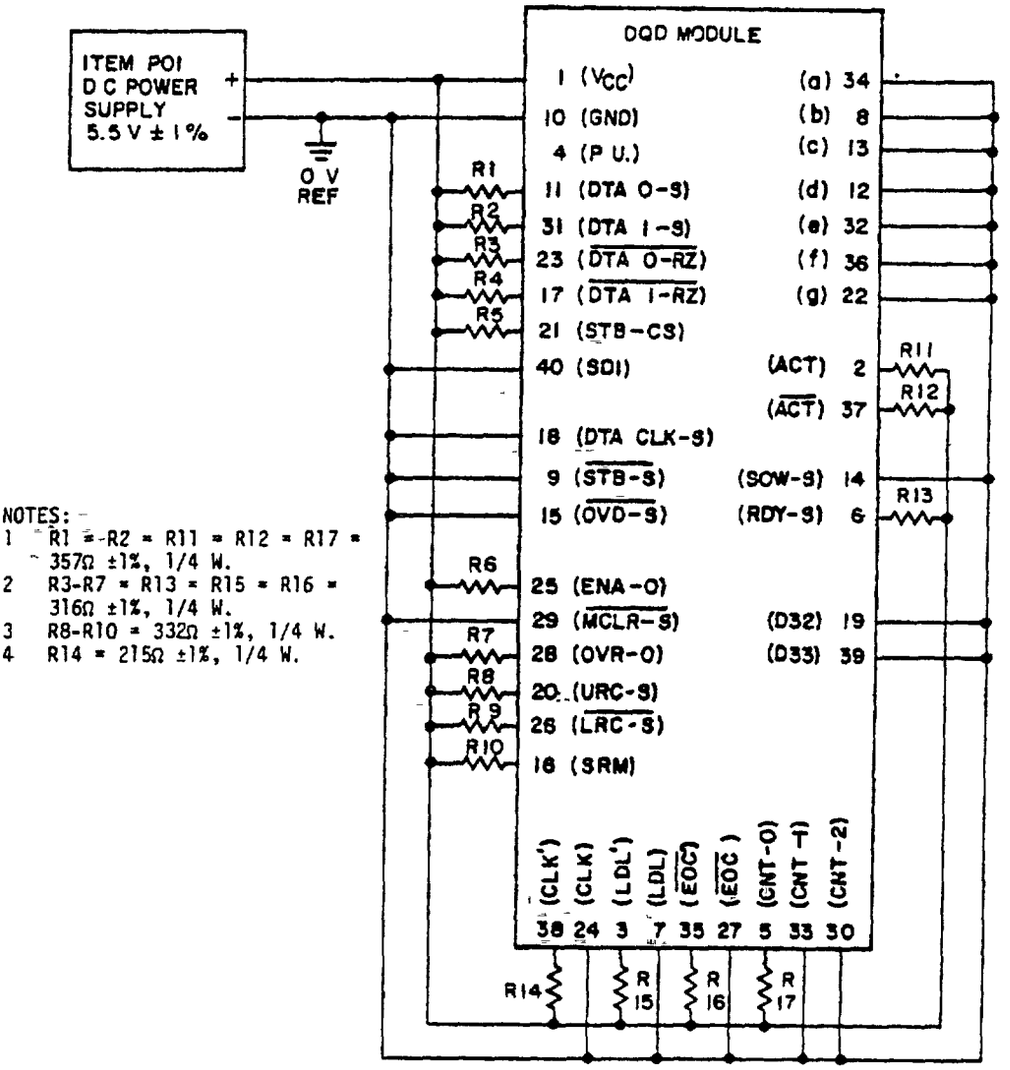



NOTE Input pulse parameters:
PRP = 1,000 ±200 ns
PW = 500 ±50 ns.
$t_{TLH} = t_{THL} \leq 10$ ns.

FIGURE 6 Operating temperature and life test circuit waveform



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NOTES:
1. R1 = R2 = R11 = R12 = R17 = 357Ω ±1%, 1/4 W.
2. R3-R7 = R13 = R15 = R16 = 316Ω ±1%, 1/4 W.
3. R8-R10 = 332Ω ±1%, 1/4 W.
4. R14 = 215Ω ±1%, 1/4 W.

FIGURE 7 Component temperature test circuit



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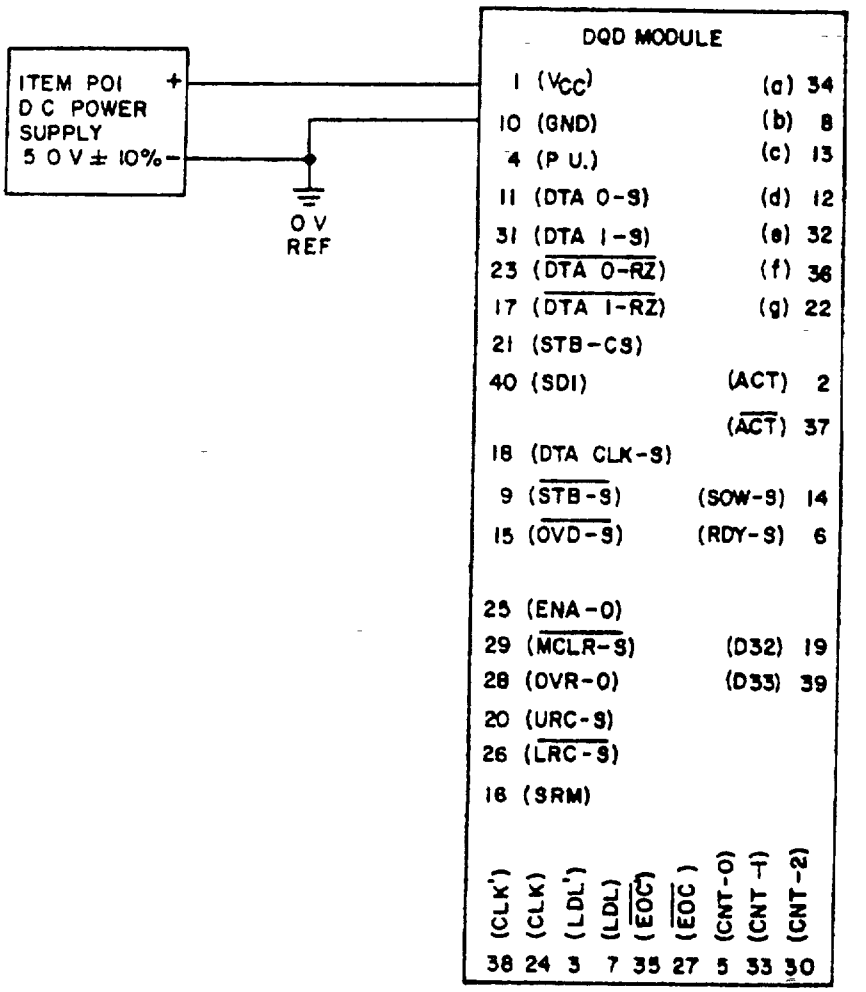



FIGURE 8 Vibration test circuit



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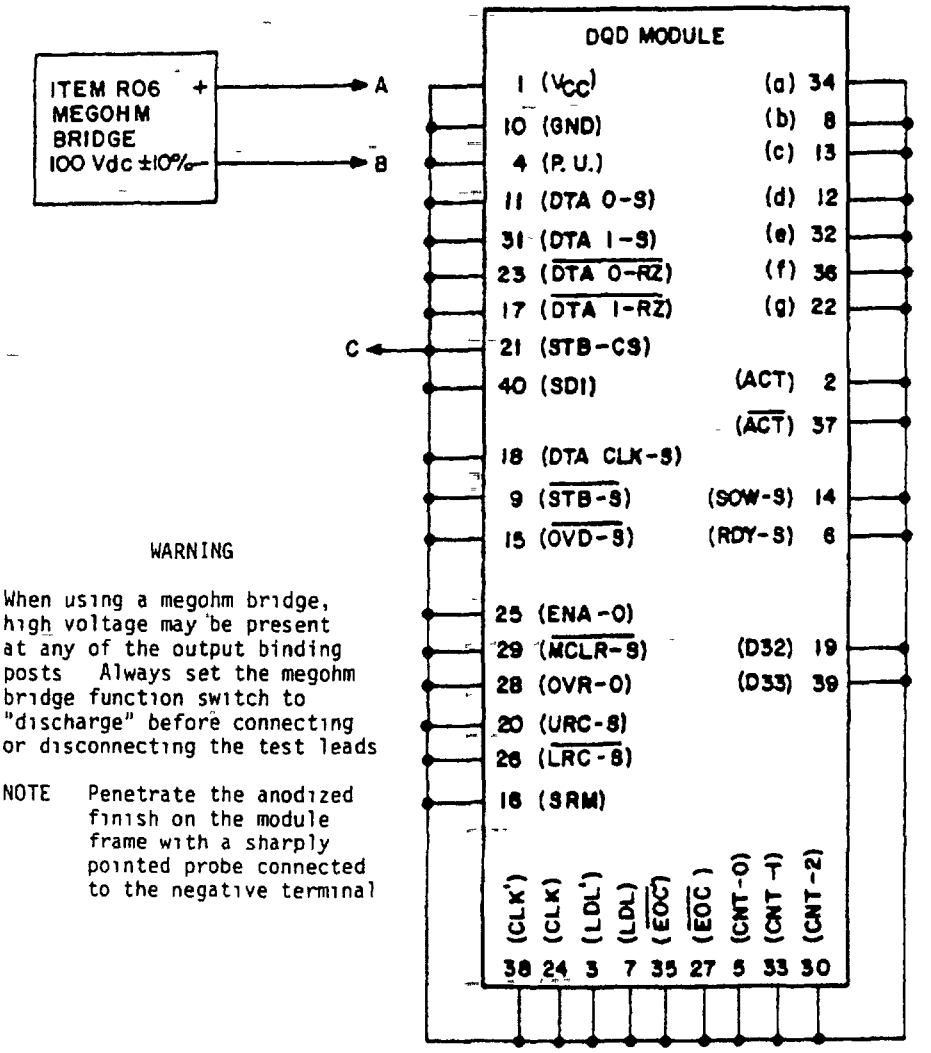
TABLE III. Isolation test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A | Isolation | C | Frame | 10 | ∞ | MΩ | 9 |





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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 9 Isolation test circuit for table III



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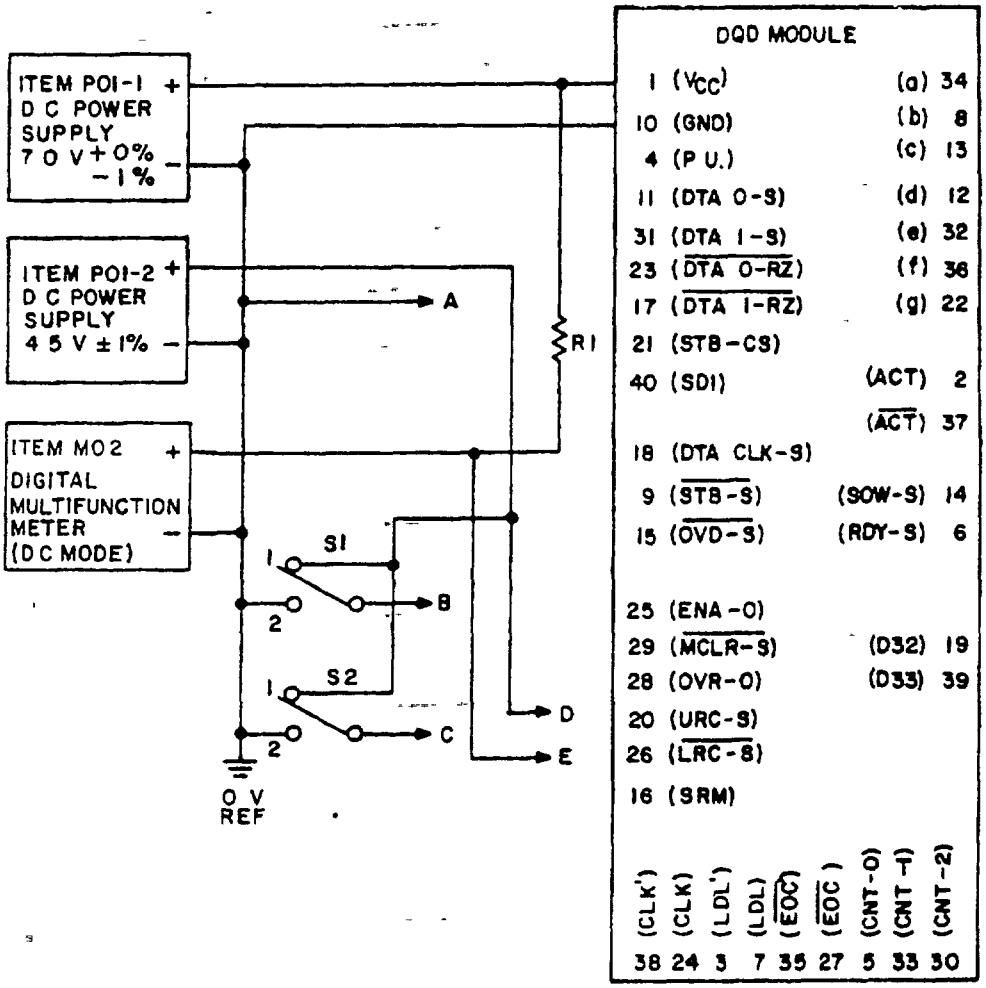
TABLE IV. $V_{OUT}$ (low-loaded) test.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| B1 | DTA 1-S $V_{OUT}$ (low loaded) | 4,29 | Open | Open | Open | 31 | 0.00 | 0.80 | V | 10 |
| B2 | $\overline{\text{DTA 0-RZ}}$ | 4,24 | | | 15,18, 29 | 23 | | | | |
| B3 | $\overline{\text{DTA 1-RZ}}$ | 4,24 | | | 15,18, 29 | 17 | | | | |
| B4 | STB-CS | 18 | | | Open | 21 | | | | |
| B5 | OVR-O | 18 | | | Open | 28 | | | | |
| B6 | URC-S | Open | | | 9 | 20 | | | | |
| B7 | $\overline{\text{LRC-S}}$ | Open | | | 29,15, 7,18 | 26 | | | | |
| B8 | SRM | Open | | | 7 | 16 | | | | |
| B9 | CLK' | 18 | | | Open | 38 | | | | |
| B10 | IDL' | 29 | | | 15,34, 8,13 | 3 | | | | |
| B11 | $\overline{\text{FOC}}$' | 29 | | | 12,32, 36,22 | 35 | | | | |
| B12 | CNT-O | 14 | 29 | 24 | 7,15, 18,27 | 5 | | | | |
| B13 | ACT | 29 | Open | Open | Open | 2 | | | | |
| B14 | $\overline{\text{ACT}}$ | 4 | Open | Open | Open | 37 | | | | |
| B15 | RDY-S | 9 | Open | Open | Open | 6 | | | | |
| B16 | DTA O-S | Open | 9 | 29 | 39,19, 7,18,15 | 11 | | | | |





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NOTES
1 For test conditions B1, B12-B14, and B16 R1 = 357Ω ±1%, 1/4 W
2 For test conditions B2-B5, B10, B11 and B15 R1 = 316Ω ±1%, 1/4 W
3 For test conditions B6-B8 R1 = 332Ω ±1%, 1/4 W
4 For test condition B9 R1 = 215Ω ±1%, 1/4 W
5 S1 = S2 = SPDT switch
6 If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating
7 When points B and C are connected to the module pins, the following switching sequence must be performed before making the measurement Switch S1 from position 1 to position 2 for a minimum of 100 ns and return it to position 1 Then switch S2 from position 1 to position 2 for a minimum of 100 ns and return it to position 1
9 After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded

FIGURE 10. $V_{OUT}$ (low-loaded) test circuit for table IV.



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TABLE V. Input breakdown current test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C1 | D32 Input breakdown current | 39 | 19 | 0.0 | 1.0 | mA | 11 |
| C2 | D33 | 19 | 39 | | | | |
| C3 | b | Open | 8 | | | | |
| C4 | d | Open | 12 | | | | |
| C5 | c | Open | 13 | | | | |
| C6 | SOW-S | 7 | 14 | | | | |
| C7 | g | Open | 22 | | | | |
| C8 | $\overline{\text{MCLR-S}}$ | | 29 | | | | |
| C9 | CNT-2 | | 30 | | | | |
| C10 | e | | 32 | | | | |
| C11 | CNT-1 | | 33 | | | | |
| C12 | a | | 34 | | | | |
| C13 | f | | 36 | | | | |
| C14 | DTA CLK-S | 29 | 18 | | 2.0 | | |
| C15 | ENA-0 | Open | 25 | | | | |
| C16 | $\overline{\text{EOC}}$ | 29 | 27 | | | | |
| C17 | $\overline{\text{STB-S}}$ | Open | 9 | | | | |
| C18 | $\overline{\text{OVD-S}}$ | 29 | 15 | | | | |





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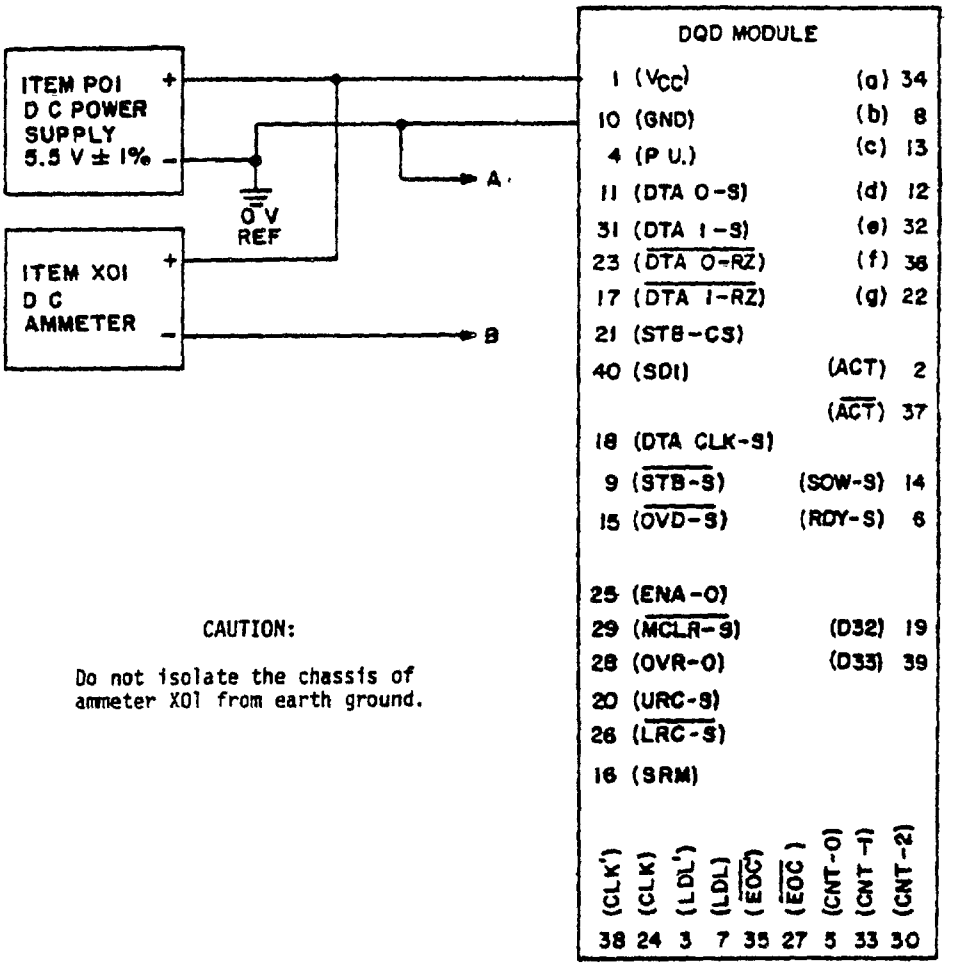
TABLE V. Input breakdown current test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| C19 | SDI Input breakdown current | Open | 40 | 0.0 | 0.2 | mA | 11 |
| C20 | LDL Input breakdown current | 14,18,29 | 7 | 0.0 | 9.0 | mA | 11 |
| C21 | CLK Input breakdown current | Open | 24 | 0.0 | 8.0 | mA | 11 |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 11 Input breakdown current test circuit for table V.



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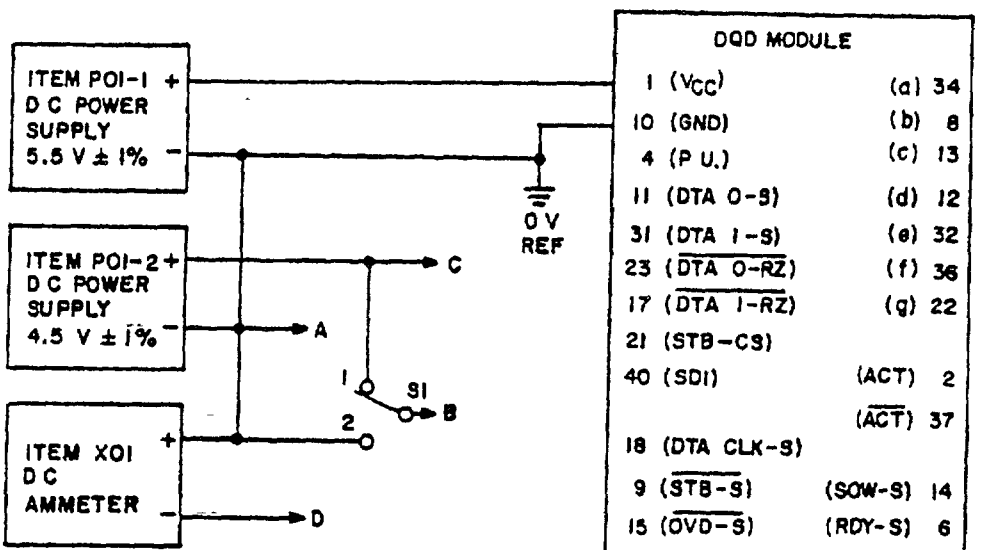
TABLE VI. Output short circuit current test.

| Test condition | Parameter | Connect | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| D1 | ACT Output short circuit current | 4 | Open | Open | 2 | -20 | -60 | mA | 12 |
| D2 | LDL' | 4,27 | | 29 | 3 | | | | |
| D3 | CNT-O | 18 | | Open | 5 | | | | |
| D4 | RDY-S | 29 | | | 6 | | | | |
| D5 | DTA O-S | 4 | | | 11 | | | | |
| D6 | SRM | 7 | | | 16 | | | | |
| D7 | $\overline{\text{DTA 1-RZ}}$ | 18 | | | 17 | | | | |
| D8 | URC-S | 9 | | | 20 | | | | |
| D9 | STB-CS | 4 | 29 | 18,27, 15 | 21 | | | | |
| D10 | $\overline{\text{DTA O-RZ}}$ | 18 | Open | Open | 23 | | | | |
| D11 | $\overline{\text{LRC-S}}$ | 18 | Open | Open | 26 | | | | |
| D12 | OVR-O | Open | 15 | 7,18,29 | 28 | | | | |
| D13 | DTA 1-S | 4 | Open | 29 | 31 | | | | |
| D14 | $\overline{\text{EOC}}$' | 29,34 | | Open | 35 | | | | |
| D15 | $\overline{\text{ACT}}$ | 29 | | Open | 37 | | | | |
| D16 | CLK' | Open | | 15,29, 18 | 38 | -40 | -120 | | |





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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. S1 = SPDT switch.
2. When point B is connected to a module pin the following switching sequence must be performed before making the measurement. Switch S1 from position 1 to 2 for a minimum of 100 ns and return it to position 1

FIGURE 12 Output short circuit current test circuit for table VI



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TABLE VII. Functional test

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| | Functional | A | A | A | B | A | A | B | B | B | B | A | B | A | A | B | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | 13 |
| E1 | | B | A | A | B | A | A | B | B | B | B | A | B | A | A | B | A | B | A | A | A | B | B | H | L | H | H | H | L | H | H | L | H | H | L | L | H | H | L | |
| E2 | | B | A | A | B | A | A | B | A | B | B | A | B | A | A | B | A | B | A | A | A | B | B | H | L | H | H | H | L | H | H | L | H | H | L | L | H | H | L | |
| E3 | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | B | A | B | A | A | A | B | B | L | L | H | H | H | L | H | H | L | H | H | L | L | H | H | L | |
| E4 | | A | A | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | H | L | H | L | L | L | H | H | H | |
| | | A | A | A | A | A | A | B | B | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E5 | | A | A | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | L | L | H | L | H | L | H | H | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | B | B | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E6 | | A | A | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | H | H | L | H | L | L | H | L | L | L | H | H | H | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | Y | X | X | X | |
| E7 | | B | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | H | H | H | L | H | L | L | H | H | L | H | H | L | L | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | B | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E8 | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | A | A | A | A | B | B | L | L | H | H | H | L | H | L | L | H | H | L | L | L | H | L | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | Y | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E9 | | A | B | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | L | L | H | H | H | H | H | L | L | H | H | L | L | L | H | H | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | B | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | Y | X | X | X | |
| E10 | | A | B | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | L | L | H | H | H | H | H | L | L | H | H | L | L | L | H | H | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | A | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.



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TABLE VII Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E11 | Functional | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | A | A | A | A | B | B | L | L | H | H | H | H | H | L | L | H | H | L | L | L | H | H | 13 |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | A | B | A | B | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | B | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | B | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E12 | | A | A | A | A | A | A | B | A | A | B | A | B | B | B | A | A | A | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | L | H | | |
| E13 | | A | A | A | A | B | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| E14 | | A | A | A | A | A | B | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| E15 | | A | A | B | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| E16 | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | B | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | B | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E17 | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | L | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | B | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | B | A | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | B | B | B | A | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | B | B | B | A | A | B | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E18 | | A | A | A | A | A | A | B | A | A | B | B | B | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | B | B | B | B | A | A | A | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | B | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.





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TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| L19 | Functional | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | 13 |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | B | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | B | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L20 | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | B | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L21 | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | L | L | H | |
| | | A | A | B | A | B | B | B | A | A | B | A | B | B | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | B | B | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | A | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L22 | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | A | B | B | A | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | A | B | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | B | A | B | B | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L23 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | B | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L24 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | B | A | A | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | B | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | B | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | B | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table



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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | P n 4 to 7 | P i n 8 to 9 | P i n 12 to 13 | P i n 14 to 15 | P i n 18 to 19 | P i n 22 to 24 | P i n 25 to 27 | P i n 29 to 30 | P i n 32 to 33 | P i n 34 to 36 | P i n 39 to 40 | P n 2 | P i n 3 | P i n 5 to 6 | P i n 11 | P i n 16 to 17 | P i n 20 to 21 | P i n 23 to 26 | P i n 28 to 31 | P i n 35 to 37 | P i n 38 | |
| | Functional | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E25 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E26 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E27 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E28 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E29 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E30 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E31 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| P32 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E33 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E34 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E35 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |
| E36 | | A A | A A | A A | B A | A A | A A | B A | A A | A A | A A | B B | B B | H L | L H | H H | L H | L L | L L | L L | L L | L L | H H | |

See footnotes at end of table



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TABLE VII Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E37 | Functional | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | 13 |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | B | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E38 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | B | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E39 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | B | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E40 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | B | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E41 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | L | H | |
| E42 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E43 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E44 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E45 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E46 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E47 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E48 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E49 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | B | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.



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TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E50 | Functional | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | H | L | H | 1 |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | |
| E51 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E52 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E53 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E54 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E55 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E56 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | H | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | B | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E57 | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | A | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | L | H | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E58 | | A | A | A | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | L | H | H | L | H | L | L | L | L | L | L | L | H | | | |
| E59 | | A | A | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | L | L | H | L | H | H | L | L | L | L | L | L | H | | | |
| E60 | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | H | L | H | L | H | L | L | H | H | L | L | L | L | | | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| E61 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | | | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | | | |

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TABLE VII Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E62 | Functional | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | 13 |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E63 | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | L | H | H | H | L | H | H | L | H | L | L | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E64 | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | L | L | H | H | L | H | H | L | H | L | L | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E65 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| I66 | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | L | L | H | H | L | H | H | L | H | L | L | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| I67 | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E68 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | B | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.



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TABLE VII. Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 1 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E69 | Functional | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | H | L | H | L | H | L | H | H | L | H | H | L | L | L | L | L | 13 |
| | | A | A | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E70 | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | L | H | H | H | L | H | H | L | H | L | L | L | |
| | | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E71 | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | H | L | H | H | L | H | H | L | L | L | L | L | |
| | | A | A | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E72 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | L | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E73 | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | L | L | H | H | L | H | H | L | H | L | L | L | |
| | | A | A | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E74 | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | H | L | H | L | H | L | H | H | L | H | H | L | L | L | L | L | |
| | | A | A | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E75 | | A | | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | H | L | H | L | L | H | H | H | L | H | H | L | H | L | L | L | |
| | | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | B | A | A | A | A | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.





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TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E76 | Functional | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | H | L | L | H | H | H | L | H | H | L | H | L | L | L | 13 |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E77 | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | L | H | H | L | H | H | L | L | L | L | L | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E78 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | I | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E79 | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | A | H | L | H | L | H | H | H | H | L | H | H | L | L | L | L | L | |
| | | A | A | A | A | A | A | B | A | B | B | A | A | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | A | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | B | A | A | A | B | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | A | A | A | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E80 | | A | A | A | A | A | A | B | A | B | A | A | A | A | A | A | A | B | A | A | A | A | A | H | L | H | L | L | L | H | L | L | H | H | L | H | H | L | L | |
| | | A | A | A | A | A | A | B | A | B | A | A | A | A | A | B | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | B | A | A | A | A | A | B | A | B | A | A | A | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E81 | | A | A | A | A | A | A | B | A | B | A | A | A | A | A | A | A | B | A | A | A | A | A | H | L | H | H | H | L | H | L | L | H | H | L | H | H | L | L | |
| | | A | A | A | B | A | A | B | A | B | A | A | A | A | A | B | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | A | A | A | A | A | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E82 | | A | B | A | B | A | A | B | A | B | A | A | B | A | B | A | A | B | A | A | A | A | A | H | H | H | L | H | H | H | H | L | H | H | L | L | H | L | L | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | B | A | B | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | A | A | B | A | B | A | B | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | B | A | A | B | A | A | B | A | B | A | A | A | A | A | A | B | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | B | A | B | A | A | A | A | A | A | B | B | A | A | A | A | A | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.



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TABLE VII. Functional test - Continued.

| Test condition letter | Test level | Connect | Output logic states (see footnotes) | Test figure |
|---|---|---|---|---|
| | | P 1 1 1

TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E91 | Functional | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | L | L | H | L | L | H | H | H | 13 |
| | | A | A | A | A | A | A | B | A | A | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E92 | | A | B | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | H | H | H | H | H | L | L | H | H | L | L | H | L | L |
| | | A | B | A | A | A | A | A | A | B | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E93 | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | L | H | H | H | H | L | H | L | L | L | H | L | H | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E94 | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | H | L | H | H | H | L | H | L | H | L | L | L | H | L | H | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E95 | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | H | H | H | H | L | L | L | L | H | H | H | L | H | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E96 | | A | B | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | L | L | H | H | H | H | H | L | H | H | H | L | L | H | H | |
| | | A | B | A | A | A | A | A | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E97 | | A | B | A | B | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | L | H | H | H | H | L | H | H | L | L | H | H | |
| | | A | B | A | B | A | A | B | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E98 | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | L | H | H | L | H | L | H | I | H | H | L | L | L | H | H | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

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TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E99 | Functional | A | A | A | A | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | H | H | L | H | L | L | H | H | L | H | H | L | L | 13 |
| | | A | A | A | A | A | A | A | A | B | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E100 | | A | A | A | B | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | H | L | H | L | L | H | H | L | L | L | H | H | L | H | |
| | | A | A | A | B | A | A | A | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E101 | | A | B | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | L | H | H | L | L | L | L | H | L | H | H | L | H | |
| | | A | B | A | A | A | A | B | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E102 | | A | B | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | L | H | H | L | H | H | H | L | H | H | H | L | L | H | H | H | |
| | | A | B | A | A | A | A | A | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| F103 | | A | A | A | A | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | L | L | H | H | L | L | H | H | L | |
| | | A | A | A | A | A | A | A | A | B | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E104 | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | H | H | H | L | L | L | H | L | L | L | H | H | L | H | |
| | | A | A | A | A | A | A | A | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E105 | | A | B | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | H | L | H | L | H | H | L | H | H | L | H | L | H | H | L | H | |
| | | A | B | A | B | A | A | B | A | A | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E106 | | A | A | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | H | L | H | L | H | L | L | H | H | L | L | H | L | L | |
| | | A | A | A | A | A | A | A | A | B | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table.



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TABLE VII Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| L107 | Functional | A | B | A | B | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | L | H | H | H | H | L | H | H | L | H | H | L | L | 1 |
| | | A | B | A | B | A | A | A | A | B | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | A | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| F108 | | A | A | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | L | H | H | L | H | L | H | L | L | H | H | L | L | H | H | L | |
| | | A | A | A | A | A | A | A | A | B | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| I109 | | A | B | A | B | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | L | H | H | L | H | L | L | H | L | H | H | L | H | |
| | | A | B | A | B | A | A | B | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| I110 | | A | A | A | A | A | A | A | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | L | H | H | L | H | L | H | L | L | H | H | L | L | H | H | L | |
| | | A | A | A | A | A | A | A | A | B | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| F111 | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | H | L | H | H | H | I | L | L | H | L | L | L | H | H | L | H | |
| I112 | | A | A | A | A | A | A | A | B | A | B | A | B | B | B | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | L | L | H | H | L | H | H | H | L | |
| I113 | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | B | A | B | A | A | A | B | B | L | L | H | H | H | L | H | L | L | H | H | L | L | H | H | L | |
| I114 | | A | A | A | A | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | L | L | H | H | H | L | H | L | L | H | L | L | L | H | H | H | |
| I115 | | A | A | A | B | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | L | I | H | L | H | L | H | H | L | H | L | L | L | H | H | H | |
| | | A | A | A | B | A | A | A | A | A | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| L116 | | A | A | A | B | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | H | L | H | L | H | L | L | H | L | L | L | L | H | H | L | H | |
| | | A | A | A | B | A | A | A | A | A | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| I117 | | A | A | A | B | A | A | A | A | A | B | A | B | B | B | A | A | B | A | A | A | B | B | H | L | H | L | H | L | I | H | L | L | L | L | H | H | L | H | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | A | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |

See footnotes at end of table



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TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | | | |

TABLE VII Functional test - Continued

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E127 | Functional | A | A | A | B | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | L | H | L | L | H | H | L | L | H | H | H | L | H | 13 |
| | | A | A | A | B | A | A | A | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E128 | | A | A | A | B | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | H | H | L | H | L | L | H | H | L | L | H | H | H | L | H | |
| | | A | A | A | B | A | A | A | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | B | A | A | A | A | A | B | A | A | A | B | B | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E129 | | A | A | A | B | A | A | A | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | L | L | H | L | H | L | H | H | L | H | L | L | L | H | H | H | |
| E130 | | A | A | A | B | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | H | H | L | H | L | L | H | H | L | L | L | H | H | L | H | |
| E131 | | A | A | A | A | A | A | A | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | H | H | H | L | H | L | L | L | H | L | L | L | H | H | L | H | |
| E132 | | A | A | A | A | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | H | L | H | H | H | L | L | L | H | L | L | L | H | H | L | H | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | B | A | B | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | B | A | B | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | A | A | A | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E133 | | A | A | A | A | A | A | A | A | A | B | A | B | B | A | A | A | B | A | A | A | B | B | H | L | H | H | H | L | L | L | H | L | L | L | H | H | L | H | |
| | | A | A | A | B | A | A | A | A | B | B | A | B | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X

TABLE VII. Functional test - Continued

| Test condition | Parameter | Connect | | |

TABLE VII Functional test - Continued.

| Test condition | Parameter | Connect | | | | | | | | | | | | | | | | | | | | | | Output logic states (see footnotes) | | | | | | | | | | | | | | | | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 4 to | Pin 7 to | Pin 8 to | Pin 9 to | Pin 12 to | Pin 13 to | Pin 14 to | Pin 15 to | Pin 18 to | Pin 19 to | Pin 22 to | Pin 24 to | Pin 25 to | Pin 27 to | Pin 29 to | Pin 30 to | Pin 32 to | Pin 33 to | Pin 34 to | Pin 36 to | Pin 39 to | Pin 40 to | Pin 2 | Pin 3 | Pin 5 | Pin 6 | Pin 11 | Pin 16 | Pin 17 | Pin 20 | Pin 21 | Pin 23 | Pin 26 | Pin 28 | Pin 31 | Pin 35 | Pin 37 | Pin 38 | |
| E142 | Functional | A | A | A | A | A | A | A | A | A | B | A | A | B | A | A | A | B | A | A | A | B | B | H | L | L | L | L | L | L | L | H | H | L | H | H | H | L | H | 13 |
| | | A | A | A | A | A | A | A | A | B | B | A | A | B | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | B | B | B | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | B | A | A | B | A | B | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E143 | | A | B | A | A | A | A | B | A | B | B | A | B | B | B | A | A | B | A | A | A | B | B | L | H | H | L | L | H | H | L | L | H | H | L | H | H | H | L | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | B | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | B | A | A | A | A | B | A | B | B | A | A | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| F144 | | A | B | A | A | A | A | B | A | B | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | L | L | H | H | L | L | H | H | L | H | H | H | L | |
| | | A | B | A | A | A | A | B | B | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| F145 | | A | B | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | L | L | H | L | L | H | H | L | L | H | H | L | H | H | H | H | |
| | | A | B | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | B | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | B | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| | | A | A | A | A | A | A | B | A | A | B | A | B | A | A | A | A | B | A | A | A | B | B | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | |
| E146 | | A | A | A | A | A | A | B | A | A | B | A | A | B | B | A | A | B | A | A | A | B | B | H | L | H | H | H | L | H | L | L | L | L | L | L | H | L | H | |

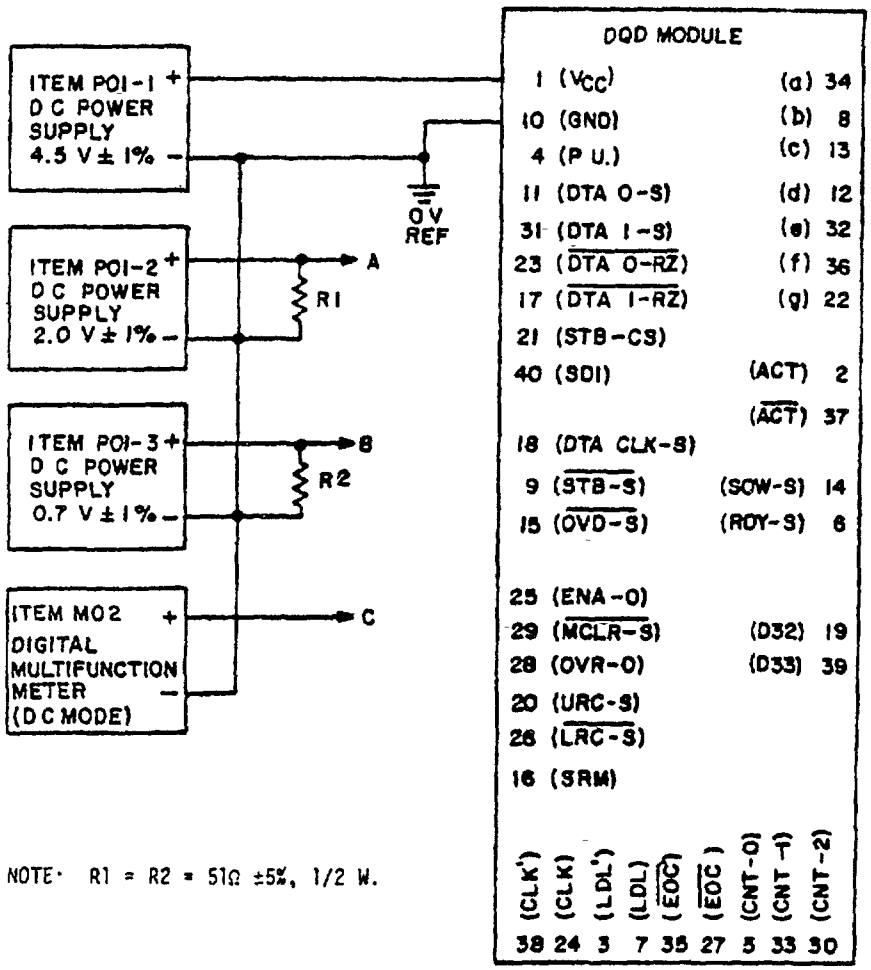
1/ Initial and end-of-life limits at 25° ±5°C ambient and 0° to 60°C operating temperature:

L = 0.0 V minimum, 0.4 V maximum
H = 2.4 V minimum, 4.5 V maximum
X = Don't care

2/ Verification of output logic states may be performed using a logic state indicator in place of item M02 shown on figure 13



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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 13. Functional test circuit for table VII



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TABLE VIII. Input clamp diode voltage test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| F1 | IDL | Input clamp diode voltage | C | 7 | -0.4 | -1.5 | V | 14 |
| F2 | b | | | 8 | | | | |
| F3 | $\overline{\text{STB-S}}$ | | | 9 | | | | |
| F4 | d | | | 12 | | | | |
| F5 | c | | | 13 | | | | |
| F6 | SOW-S | | | 14 | | | | |
| F7 | $\overline{\text{OVD-S}}$ | | | 15 | | | | |
| F8 | DTA CLK-S | | | 18 | | | | |
| F9 | D32 | | D | 19 | | | | |
| F10 | g | | C | 22 | | | | |
| F11 | CLK | | | 24 | -0.1 | | | |
| F12 | ENA-O | | | 25 | -0.4 | | | |
| F13 | $\overline{\text{EOC}}$ | | | 27 | | | | |
| F14 | $\overline{\text{MCLR-S}}$ | | | 29 | | | | |
| F15 | CNT-2 | | | 30 | | | | |
| F16 | e | | | 32 | | | | |
| F17 | CNT-1 | | | 33 | | | | |
| F18 | a | | | 34 | | | | |





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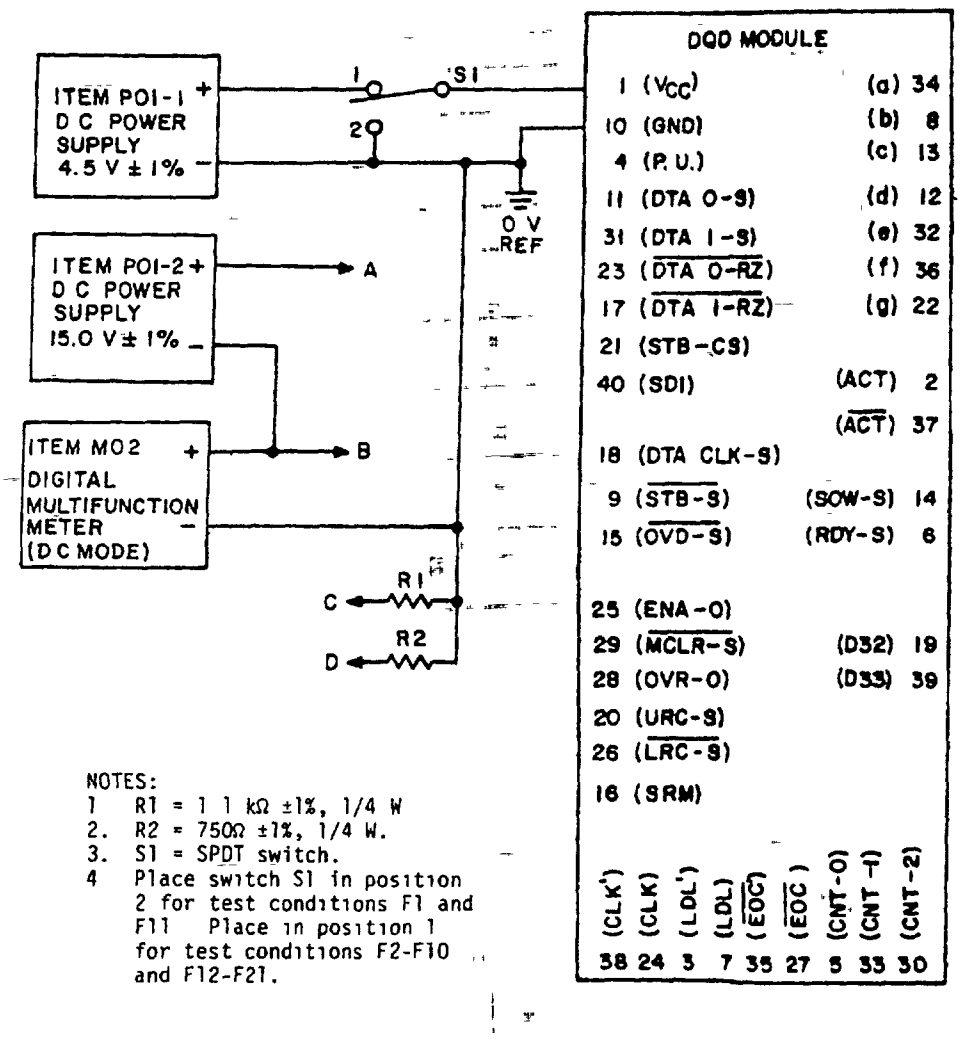
TABLE VIII. Input clamp diode voltage test - Continued.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| F19 | f Input clamp diode voltage | C | 36 | -0.4 | -1.5 | V | 14 |
| F20 | D33 Input clamp diode voltage | D | 39 | -0.4 | -1.5 | V | 14 |
| F21 | SDI Input clamp diode voltage | D | 40 | -0.4 | -1.5 | V | 14 |





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NOTES:
1. R1 = 1 1 kΩ ±1%, 1/4 W
2. R2 = 750Ω ±1%, 1/4 W.
3. S1 = SPDT switch.
4. Place switch S1 in position 2 for test conditions F1 and F11. Place in position 1 for test conditions F2-F10 and F12-F21.

FIGURE 14 Input clamp diode voltage test circuit for table VIII



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TABLE IX. Static output voltage (low-level) test.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| G1 | ACT — Static output voltage, low level | Open | 29 | 2 | Open | Open | 0.01 | 0.40 | V | 15 |
| G2 | LDL' | 15 | 29 | 3 | Open | Open | | | | |
| G3 | CNT-O | 7,15, 18,27 | 14 | 5 | 29 | 24 | | | | |
| G4 | RDY-S | Open | 9 | 6 | Open | Open | | | | |
| G5 | DATA O-S | 7,19, 39,18, 15 | Open | 11 | 9 | 29 | | | | |
| G6 | SRM | 7 | Open | 16 | Open | Open | | | | |
| G7 | $\overline{\text{DTA 1-RZ}}$ | 15,18, 29 | 4,24 | 17 | | | | | | |
| G8 | URC-S | 9 | Open | 20 | | | | | | |
| G9 | STB-CS | Open | 18 | 21 | | | | | | |
| G10 | $\overline{\text{DTA O-RZ}}$ | 15,18, 29 | 4,24 | 23 | | | | | | |
| G11 | $\overline{\text{LRC-S}}$ | 7,18, 15,29 | Open | 26 | | | | | | |
| G12 | OVR-O | Open | 18 | 28 | | | | | | |
| G13 | DTA 1-S | Open | 4,29 | 31 | | | | | | |
| G14 | $\overline{\text{EOC}}$' | 8,12, 13,22, 32,34, 36 | 29 | 35 | | | | | | |





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TABLE IX. Static output voltage (low-level) test - Continued.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| G15 | $\overline{ACT}$ Static output voltage, low level | Open | 4 | 37 | Open | Open | 0.01 | 0.40 | V | 15 |
| G16 | CLK' Static output voltage, low level | Open | 18 | 38 | Open | Open | 0.01 | 0.40 | V | 15 |





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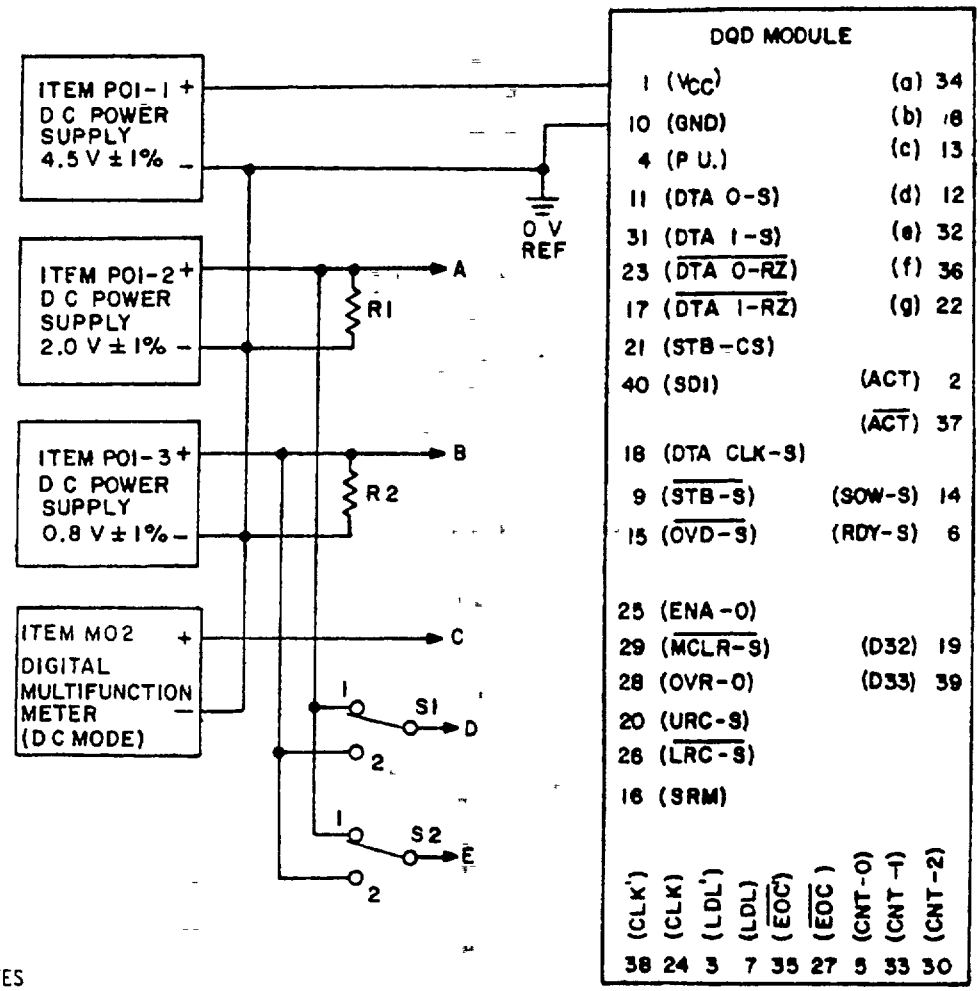
TABLE X. Static output voltage (high-level) test.

| Test condition | Parameter | Connect | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| H1 | ACT — Static output voltage, high level | Open | 4 | 2 | Open | Open | 2.4 | 4.5 | V | 15 |
| H2 | LDL' | 29 | 4,27 | 3 | | | | | | |
| H3 | CNT-O | Open | 18 | 5 | | | | | | |
| H4 | RDY-S | | 29 | 6 | | | | | | |
| H5 | DTA O-S | | 4 | 11 | | | | | | |
| H6 | SRM | | 7 | 16 | | | | | | |
| H7 | $\overline{\text{DTA 1-RZ}}$ | | 18 | 17 | | | | | | |
| H8 | URC-S | | 9 | 20 | | | | | | |
| H9 | STB-CS | 15,18, 27 | 4 | 21 | 29 | | | | | |
| H10 | $\overline{\text{DTA O-RZ}}$ | Open | 18 | 23 | Open | | | | | |
| H11 | $\overline{\text{LRC-S}}$ | Open | 18 | 26 | Open | | | | | |
| H12 | OVR-O | 7,18, 29 | Open | 28 | 15 | | | | | |
| H13 | DTA 1-S | 29 | 4 | 31 | Open | | | | | |
| H14 | $\overline{\text{EOC}}$' | Open | 29,34 | 35 | | | | | | |
| H15 | $\overline{\text{ACT}}$ | Open | 29 | 37 | | | | | | |
| H16 | CLK' | 15,18, 29 | Open | 38 | | | | | | |





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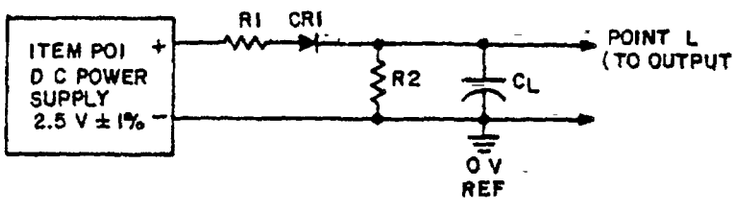



NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. S1 = S2 = SPDT switch
3. Load all outputs as specified on figure 16
4. When point D is connected to a module pin the following switching sequence must be performed before making measurement. Switch S1 from position 1 to 2 for a minimum of 100 ns and return it to position 1
5. When points D and E are connected to module pins the following switching sequence must be performed before making measurement. Switch S1 from position 1 to 2 for a minimum of 100 ns and return it to position 1. Then switch S2 from position 1 to 2 for a minimum of 100 ns and return it to position 1

FIGURE 15. Static output voltage (low-level and high-level) test circuit for tables IX and X



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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. For output pins 3, 6, 17, 21, 23, 28 and 35:
   R1 = Select value for a current flow of 16.0 mA ±5% into point L with point L held at 0.4 V.
3. For output pins 16, 20, and 26:
   R1 = Select value for a current flow of 15.6 mA ±5% into point L with point L held at 0.4 V.
4. For output pins 2, 5, 11, 31, and 37:
   R1 = Select value for a current flow of 14.4 mA ±5% into point L with point L held at 0.4 V.
5. For output pin 38:
   R1 = Select value for a current flow of 24.0 mA ±5% into point L with point L held at 0.4 V.
6. For output pins 6, 17, 21, 23, 28, and 35:
   R2 = 6.04 kΩ ±1%, 1/8 W.
7. For output pin 26:
   R2 = 3.09 kΩ ±1%, 1/8 W.
8. For output pins 20 and 16:
   R2 = 6.34 kΩ ±1%, 1/8 W.
9. For output pin 3:
   R2 = 3.01 kΩ ±1%, 1/8 W.
10. For output pins 2, 5, 11, 31, and 37:
    R2 = 6.65 kΩ ±1%, 1/8 W.
11. For output pin 38:
    R2 = 1.74 kΩ ±1%, 1/8 W.
12. For test conditions M1-M16, N1-N16, P1-P6, Q1-Q6, R1, R2, S1, S2, T1-T15, and U1-U16:
    $C_L$ = 50 ±5 pF (including probe and parasitic capacitance).
13. For test conditions M17-M32, N17-N32, P7-P12, Q7-Q12.
    $C_L$ = 150 ±5 pF (including probe and parasitic capacitance)
14. Capacitor $C_L$ is not required for dc electrical test conditions G1-G16 and H1-H16
15. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 16. DQD logic load circuit



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TABLE XI Input current (high-level) test

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| J1 | D32 | Input current, high level | Open | 19 | Open | 0 | 20 | μA | 17 |
| J2 | D33 | | | 39 | | | 20 | | |
| J3 | b | | | 8 | | | 40 | | |
| J4 | d | | | 12 | | | | | |
| J5 | c | | | 13 | | | | | |
| J6 | SOW-S | | | 14 | 7 | | | | |
| J7 | g | | | 22 | Open | | | | |
| J8 | $\overline{\text{MCLR-S}}$ | | | 29 | | | | | |
| J9 | CNT-2 | | | 30 | | | | | |
| J10 | e | | | 32 | | | | | |
| J11 | CNT-1 | | | 33 | | | | | |
| J12 | a | | | 34 | | | | | |
| J13 | f | | | 36 | | | | | |
| J14 | SDI | | 7 | 40 | | | | | |
| J15 | DTA CLK-S | | Open | 18 | 29, 15 | | 80 | | |
| J16 | ENA-0 | | | 25 | Open | | 80 | | |
| J17 | $\overline{\text{EOC}}$ | | | 27 | 29, 15 | | 80 | | |
| J18 | $\overline{\text{STB-S}}$ | | | 9 | 24 | | 120 | | |





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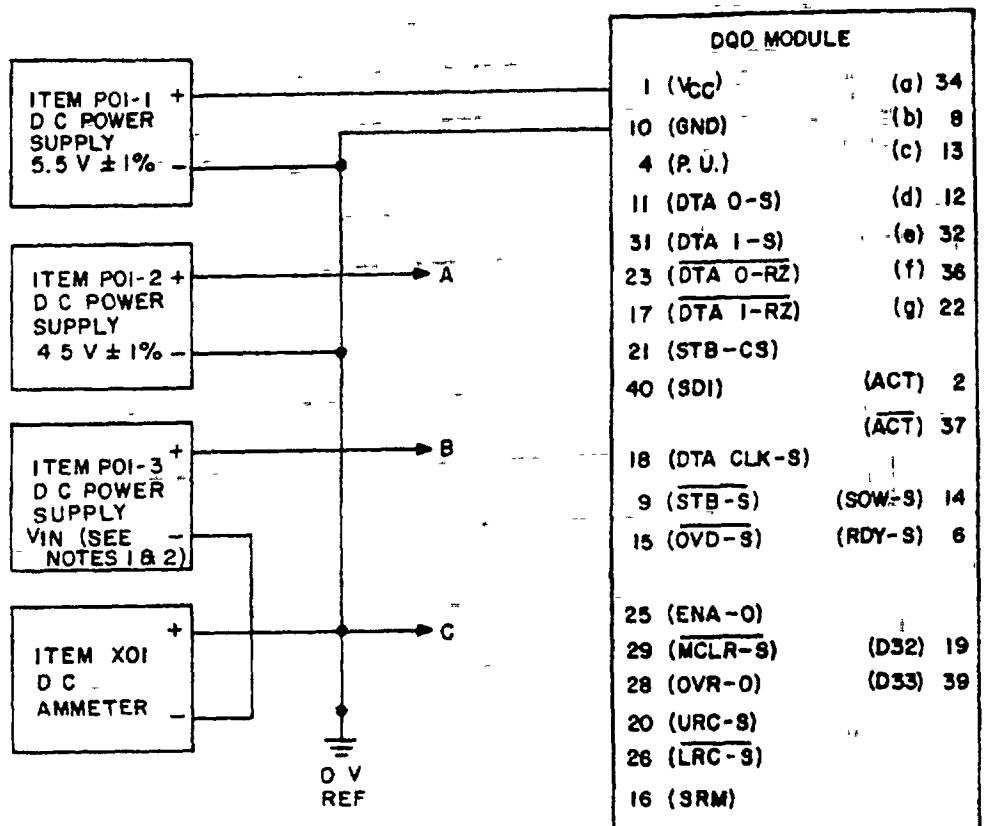
TABLE XI. Input current (high-level) test - Continued.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| J19 | $\overline{\text{OVD-S}}$ | Input current, high level | Open | 15 | 24, 29 | 0 | 120 | µA | 17 |
| J20 | LDL | Input current, high level | Open | 7 | 14,29,24 | 0 | 360 | µA | 17 |
| J21 | CLK | Input current, high level | Open | 24 | 29 | -18,000 | 640 | µA | 17 |





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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES

1. For test conditions J1, J2, and J14 $V_{IN}$ = 2.5 V ±1%.
2. For test conditions J3-J13 and J15-J21 $V_{IN}$ = 2 4 V ±1%.

FIGURE 17 Input current (high-level) test circuit for table XI



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TABLE XII. Input current (low-level) test.

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| K1 | D32 | Input current, low level | Open | Open | 19 | -0.100 | -0.389 | mA | 18 |
| K2 | D33 | | | Open | 39 | -0.100 | -0.389 | | |
| K3 | SDI | | | 7 | 40 | -0.100 | -0.778 | | |
| K4 | b | | | Open | 8 | -0.10 | -1.73 | | |
| K5 | d | | | | 12 | | | | |
| K6 | c | | | | 13 | | | | |
| K7 | SOW-S | | 7 | | 14 | | | | |
| K8 | g | | Open | | 22 | | | | |
| K9 | $\overline{\text{MCLR-S}}$ | | | | 29 | | | | |
| K10 | CNT-2 | | | | 30 | | | | |
| K11 | e | | | | 32 | | | | |
| K12 | CNT-1 | | | | 33 | | | | |
| K13 | a | | | | 34 | | | | |
| K14 | f | | | | 36 | | | | |
| K15 | $\overline{\text{STB-S}}$ | | | | 9 | | -4.80 | | |
| K16 | $\overline{\text{OVD-S}}$ | | 29 | 9 | 15 | | -4.80 | | |
| K17 | DTA CLK-S | | 15, 29 | Open | 18 | | -3.46 | | |
| K18 | ENA-0 | | Open | Open | 25 | | -3.46 | | |





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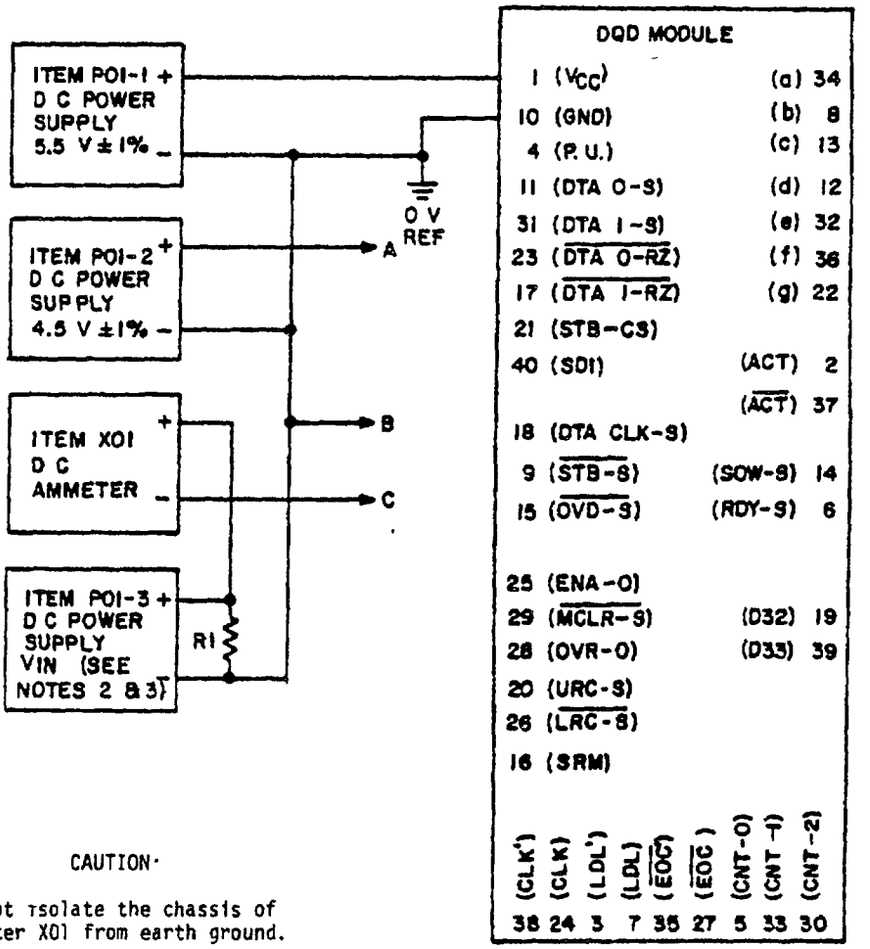
TABLE XII. Input current (low-level) test - Continued

| Test condition | Parameter | | Connect | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | Min | Max | | |
| K19 | $\overline{EOC}$ | Input current, low level | 15,18,29 | Open | 27 | -0.10 | -3.46 | mA | 18 |
| K20 | LDL | Input current, low level | 14,18,15, 29 | 4 | 7 | -0.10 | -15 6 | mA | 18 |
| K21 | CLK | Input current, low level | Open | Open | 24 | -0.10 | -25.6 | mA | 18 |





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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES
1. R1 = 10Ω ±1%, 1/4 W.
2. For test conditions K1-K14 and K17-K20: $V_{IN}$ = 0.0 ±10 mV.
3. For test conditions K15, K16, and K21: $V_{IN}$ = 0.4 V ±1%.

FIGURE 18. Input current (low-level) test circuit for table XII.



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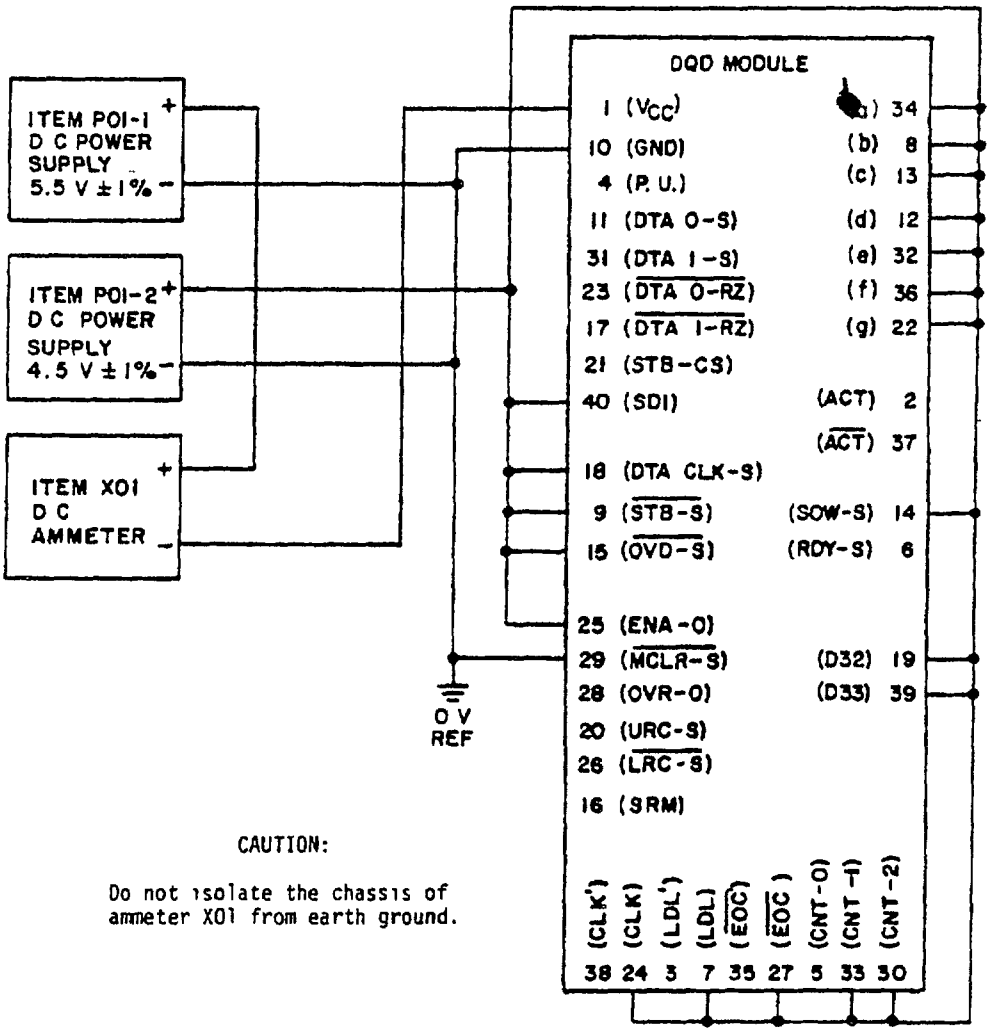
TABLE XIII. Power supply current test.

| Test condition | Parameter | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| L | Power supply current | 130 | 520 | mA | 19 |





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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 19 Power supply current test circuit for table XIII



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TABLE XIV. Transition time ($t_{TLH}$) test

| Test condition | Parameter | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | F to pin | Min | Max | | |
| M1 | ACT — Transition time ($t_{TLH}$), $C_L$ = 50 pF | Open | 24, 29 | Open | 4 | Open | 2 | 4 | 30 | ns | 20,21 |
| M2 | LDL' | 29 | 4 | Open | 27 | | 3 | | | | |
| M3 | CNT-0 | 7,15, 27,29 | 14 | 24 | 18 | | 5 | | | | |
| M4 | RDY-S | 15 | 9, 24 | Open | 29 | | 6 | | | | |
| M5 | DTA 0-S | 7,15, 19,29, 39 | 27 | 9 | 14, 18 | 24 | 11 | | | | |
| M6 | SRM | Open | Open | Open | 7 | Open | 16 | | | | |
| M7 | $\overline{\text{DTA 1-RZ}}$ | 29,15 | 4 | | 18 | | 17 | | | | |
| M8 | URC-S | Open | Open | | 9 | | 20 | | | | |
| M9 | STB-CS | 18,27, 15 | 4 | | 29 | | 21 | | | | |
| M10 | $\overline{\text{DTA 0-RZ}}$ | 15,29 | 4 | | 18 | | 23 | | | | |
| M11 | $\overline{\text{LRC-S}}$ | 18,29, 15 | 27 | | 7 | | 26 | | | | |
| M12 | OVR-0 | 29,18, 7 | Open | | 15 | | 28 | | | | |
| M13 | D1A 1-S | Open | 4 | | 29 | | 31 | | | | |
| M14 | $\overline{\text{EOC}}$' | 8,12, 13,32, 34,36 | 29 | | 22 | | 35 | | | | |





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TABLE XIV. Transition time ($t_{TLH}$) test - Continued.

| Test condition | Parameter | | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | E to pin | F to pin | Min | Max | | |
| M15 | $\overline{ACT}$ | Transition time ($t_{TLH}$), $C_L$ = 50 pF | 15 | 4 | Open | 29 | Open | 37 | 4 | 30 | ns | 20,21 |
| M16 | CLK' | Transition time ($t_{TLH}$), $C_L$ = 50 pF | 15,29 | Open | | 18 | | 38 | | 30 | | |
| M17 | ACT | Transition time ($t_{TLH}$), $C_L$ = 150 pF | Open | 24, 29 | | 4 | | 2 | | 50 | | |
| M18 | IDL' | | 29 | 4 | | 27 | | 3 | | | | |
| M19 | CNT-0 | | 7,15, 27,29 | 14 | 24 | 18 | | 5 | | | | |
| M20 | RDY-S | | 15 | 9, 24 | Open | 29 | | 6 | | | | |
| M21 | DTA 0-S | | 7,15, 19,29, 39 | 27 | 9 | 14, 18 | 24 | 11 | | | | |
| M22 | SRM | | Open | Open | Open | 7 | Open | 16 | | | | |
| M23 | $\overline{DTA\ 1-RZ}$ | | 29,15 | 4 | | 18 | | 17 | | | | |
| M24 | URC-S | | Open | Open | | 9 | | 20 | | | | |
| M25 | STB-CS | | 18,27, 15 | 4 | | 29 | | 21 | | | | |
| M26 | $\overline{DTA\ 0-RZ}$ | | 15,29 | 4 | | 18 | | 23 | | | | |
| M27 | $\overline{LRC-S}$ | | 18,29, 15 | 27 | | 7 | | 26 | | | | |
| M28 | OVR-0 | | 7,18, 29 | Open | | 15 | | 28 | | | | |





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TABLE XIV Transition time ($t_{TLH}$) test - Continued.

| Test condition | Parameter | | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | E to pin | F to pin | Min | Max | | |
| M29 | DTA 1-S | Transition time ($t_{TLH}$), $C_L$ = 150 pF | Open | 4 | Open | 29 | Open | 31 | 4 | 50 | ns | 20,21 |
| M30 | $\overline{EOC}$' | ↓ | 8,12, 13,32, 34,36 | 29 | ↓ | 22 | ↓ | 35 | ↓ | ↓ | ↓ | ↓ |
| M31 | $\overline{ACT}$ | ↓ | 15 | 4 | ↓ | 29 | ↓ | 37 | ↓ | ↓ | ↓ | ↓ |
| M32 | CLK' | ↓ | 15,29 | Open | ↓ | 18 | ↓ | 38 | ↓ | ↓ | ↓ | ↓ |





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TABLE XV Transition time ($t_{THL}$) test.

| Test condition | Parameter | | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | Connect E to pin | Connect F to pin | Initial limits 25° ±5°C ambient Min | Initial limits 25° ±5°C ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| N1 | ACT | Transition time ($t_{THL}$), $C_L$ = 50 pF | Open | 24, 29 | Open | 4 | Open | 2 | 4 | 13 | ns | 20,21 |
| N2 | IDL' | | 29 | 4 | Open | 27 | | 3 | | | | |
| N3 | CNT-0 | | 7,15, 27,29 | 14 | 24 | 18 | | 5 | | | | |
| N4 | RDY-S | | 15 | 9, 24 | Open | 29 | | 6 | | | | |
| N5 | DTA 0-S | | 7,15, 19,29, 39 | 27 | 9 | 14, 18 | 24 | 11 | | | | |
| N6 | SRM | | Open | Open | Open | 7 | Open | 16 | | | | |
| N7 | $\overline{\text{DTA 1-RZ}}$ | | 29,15 | 4 | | 18 | | 17 | | | | |
| N8 | URC-S | | Open | Open | | 9 | | 20 | | | | |
| N9 | STB-CS | | 18,27, 15 | 4 | | 29 | | 21 | | | | |
| N10 | $\overline{\text{DTA 0-RZ}}$ | | 15,29 | 4 | | 18 | | 23 | | | | |
| N11 | $\overline{\text{LRC-S}}$ | | 18,29, 15 | 27 | | 7 | | 26 | | | | |
| N12 | OVR-0 | | 7,18, 29 | Open | | 15 | | 28 | | | | |
| N13 | DTA 1-S | | Open | 4 | | 29 | | 31 | | | | |
| N14 | $\overline{\text{EOC}}$' | | 8,12, 13,32, 34,36 | 29 | | 22 | | 35 | | | | |





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TABLE XV Transition time ($t_{THL}$) test - Continued

| Test condition | Parameter | | Connect A to pin | Connect B to pin | Connect C to pin | Connect D to pin | Connect E to pin | Connect F to pin | Initial limits 25° ±5°C ambient Min | Initial limits 25° ±5°C ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| N15 | $\overline{ACT}$ | Transition time ($t_{THL}$), $C_L$ = 50 pF | 15 | 4 | Open | 29 | Open | 37 | 4 | 13 | ns | 20,21 |
| N16 | CLK' | Transition time ($t_{THL}$), $C_L$ = 50 pF | 15,29 | Open | | 18 | | 38 | | 13 | | |
| N17 | ACT | Transition time ($t_{THL}$), $C_L$ = 150 pF | Open | 24, 29 | | 4 | | 2 | | 20 | | |
| N18 | LDL' | | 29 | 4 | | 27 | | 3 | | | | |
| N19 | CNT-0 | | 7,15, 27,29 | 14 | 24 | 18 | | 5 | | | | |
| N20 | RDY-S | | 15 | 9, 24 | Open | 29 | | 6 | | | | |
| N21 | DTA 0-S | | 7,15, 19,29, 39 | 27 | 9 | 14, 18 | 24 | 11 | | | | |
| N22 | SRM | | Open | Open | Open | 7 | Open | 16 | | | | |
| N23 | $\overline{DTA\ 1-RZ}$ | | 29,15 | 4 | | 18 | | 17 | | | | |
| N24 | URC-S | | Open | Open | | 9 | | 20 | | | | |
| N25 | S1B-CS | | 18,27, 15 | 4 | | 29 | | 21 | | | | |
| N26 | $\overline{DTA\ 0-RZ}$ | | 15,29 | 4 | | 18 | | 23 | | | | |
| N27 | $\overline{LRC-S}$ | | 18,29, 15 | 27 | | 7 | | 26 | | | | |
| N28 | OVR-0 | | 7,18, 29 | Open | | 15 | | 28 | | | | |





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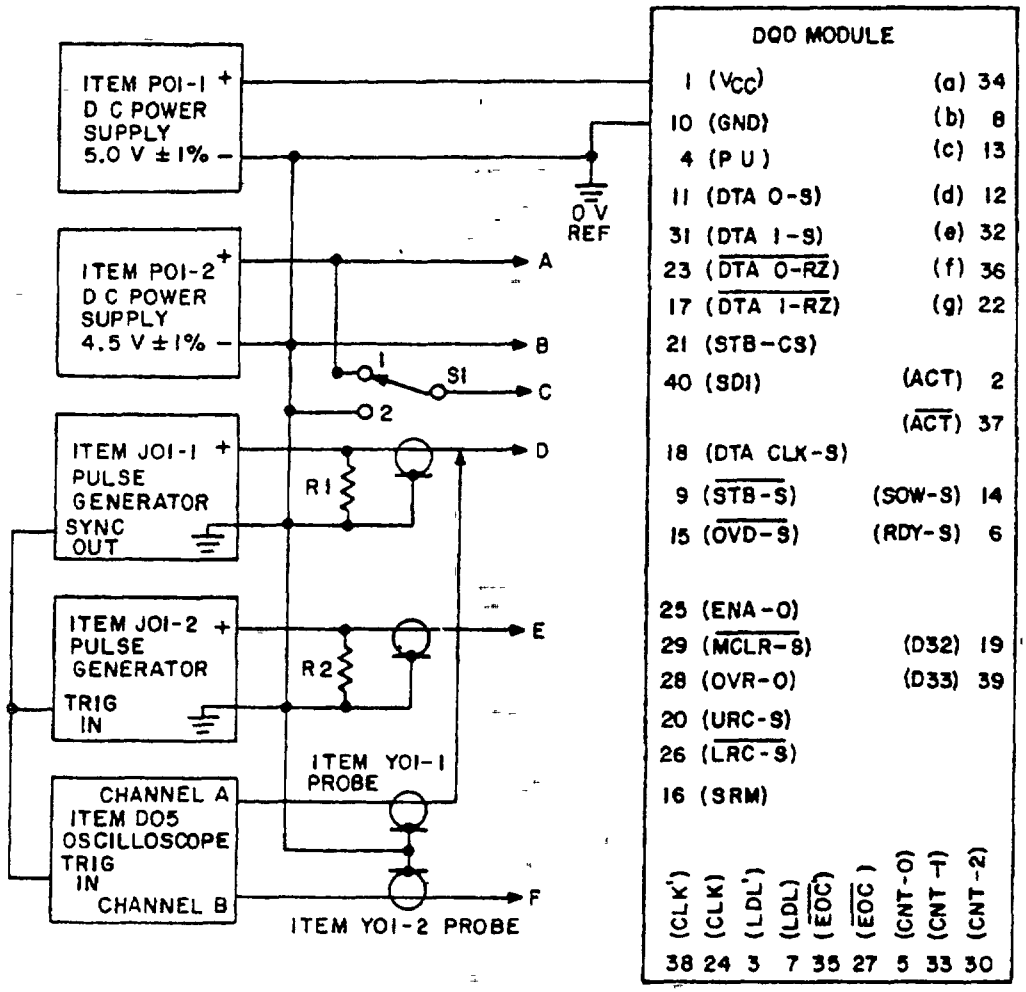
TABLE XV. Transition time ($t_{THL}$) test - Continued.

| Test condition | Parameter | | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | E to pin | F to pin | Min | Max | | |
| N29 | DTA 1-S | Transition time ($t_{THL}$), $C_L$ = 150 pF | Open | 4 | Open | 29 | Open | 31 | 4 | 20 | ns | 20,21 |
| N30 | $\overline{EOC}$' | ↓ | 8,12, 13,32, 34,36 | 29 | ↓ | 22 | ↓ | 35 | ↓ | ↓ | ↓ | ↓ |
| N31 | $\overline{ACT}$ | ↓ | 15 | 4 | ↓ | 29 | ↓ | 37 | ↓ | ↓ | ↓ | ↓ |
| N32 | CLK' | ↓ | 15,29 | Open | ↓ | 18 | ↓ | 38 | ↓ | ↓ | ↓ | ↓ |





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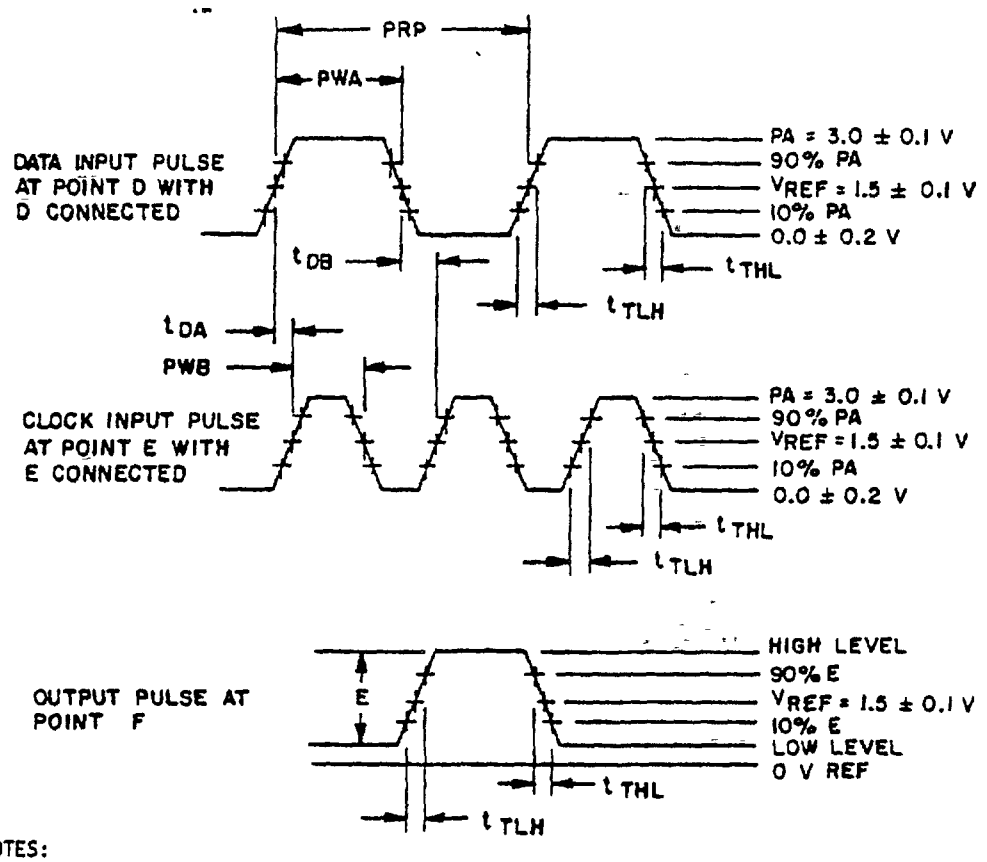



NOTES
1. R1 = R2 = 51 Ω ±1%, 1/4 W
2. S1 = SPDT switch
3. Load each output as specified on figure 16
4. When point C is connected to a module pin the following switching sequence must be performed before making the measurement. Switch S1 from position 1 to position 2 for a minimum of 100 ns and return it to position 1

FIGURE 20. Transition time ($t_{TLH}$ and $t_{THL}$) test circuit for tables XIV and XV



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NOTES:

1. Data input pulse parameters:
   PRP = 500 ±100 ns.
   PWA = 250 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Clock input pulse parameters:
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   $t_{DA}$ = 50 ±5 ns.
   $t_{DB}$ = 50 ±5 ns.
3. The clock pulse generator shall be run in the double pulse mode. The delay control on the data pulse generator controls $t_{DA}$ and the delay control on the clock pulse generator controls $t_{DB}$.
4. The time relationship between the input and output pulses is not shown.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 21. Transition time ($t_{TLH}$ and $t_{THL}$) test circuit waveforms for tables XIV and XV.



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TABLE XVI Propagation delay time ($t_{PLH}$) test.

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | \ to pin | Min | Max | | |
| P1 | DTA CLK-S to DTA 0-S | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | 11 | 20 | 122 | ns | 22,23 |
| P2 | DTA CLK-S to DTA 1-RZ | | 17 | 12 | 64 | | |
| P3 | DTA CLK-S to STB-CS | | 21 | 16 | 95 | | |
| P4 | DTA CLK-S to DTA 0-RZ | | 23 | 12 | 64 | | |
| P5 | DTA CLK-S to LRC-S | | 26 | 12 | 72 | | |
| P6 | DTA CLK-S to DTA 1-S | | 31 | 24 | 131 | | |
| P7 | DTA CLK-S to DTA 0-S | Propagation delay time ($t_{PLH}$), $C_L$ = 150 pF | 11 | 20 | 127 | | |
| P8 | DTA CLK-S to DTA 1-RZ | | 17 | 12 | 69 | | |
| P9 | DTA CLK-S to STB-CS | | 21 | 16 | 100 | | |
| P10 | DTA CLK-S to DTA 0-RZ | | 23 | 12 | 69 | | |
| P11 | DTA CLK-S to LRC-S | | 26 | 12 | 77 | | |
| P12 | DTA CLK-S to DTA 1-S | | 31 | 24 | 136 | | |





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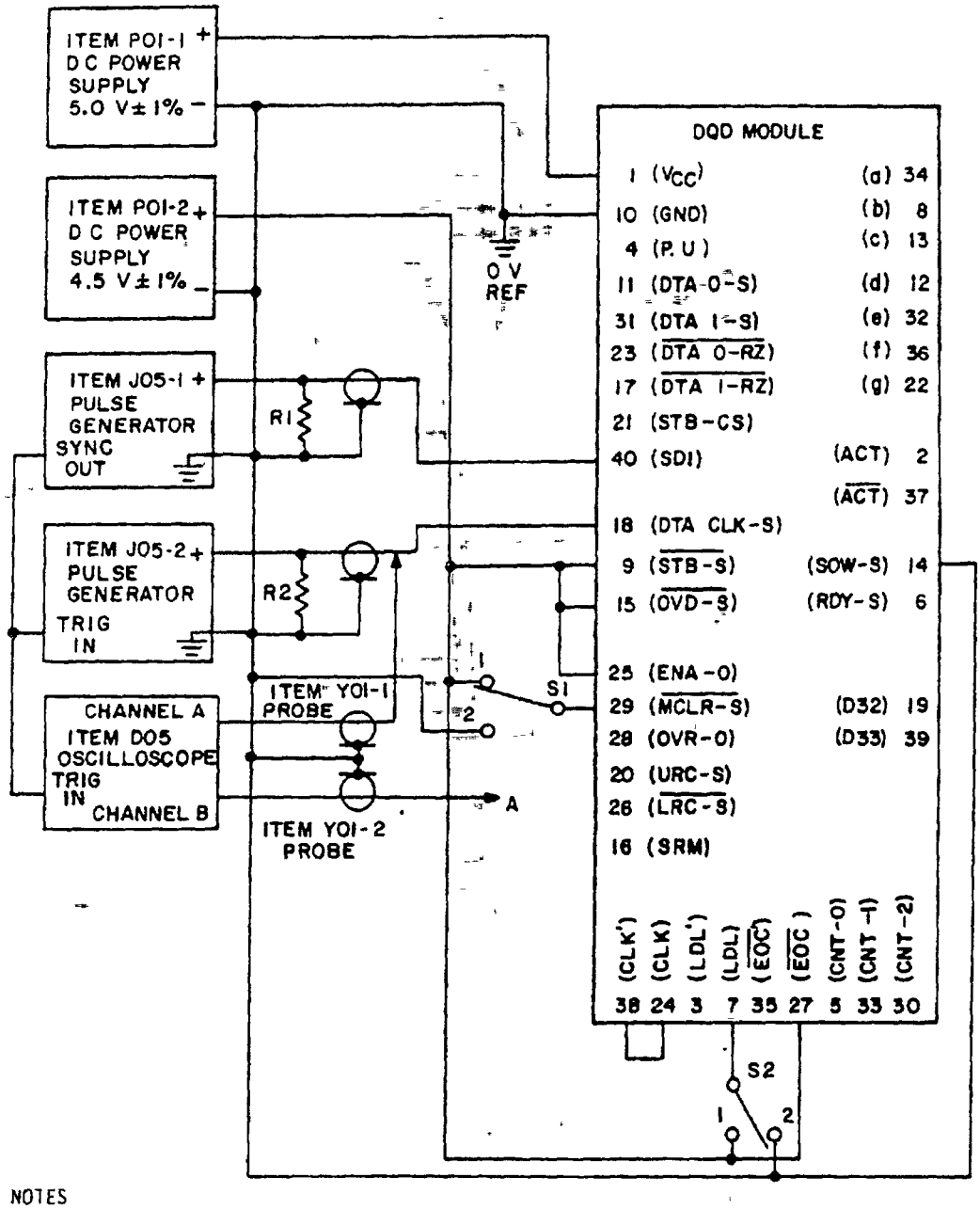
TABLE XVII. Propagation delay time ($t_{PHL}$) test.

| Test condition | Parameter | | Connect | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | | A to pin | Min | Max | | |
| Q1 | DTA CLK-S to DTA 0-S | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | 11 | 20 | 108 | ns | 22,23 |
| Q2 | DTA CLK-S to DTA 1-RZ | | 17 | 12 | 72 | | |
| Q3 | DTA CLK-S to STB-CS | | 21 | 16 | 85 | | |
| Q4 | DTA CLK-S to DTA 0-RZ | | 23 | 12 | 72 | | |
| Q5 | DTA CLK-S to LRC-S | | 26 | 12 | 64 | | |
| Q6 | DTA CLK-S to DTA 1-S | | 31 | 24 | 136 | | |
| Q7 | DTA CLK-S to DTA 0-S | Propagation delay time ($t_{PHL}$), $C_L$ = 150 pF | 11 | 20 | 115 | | |
| Q8 | DTA CLK-S to DTA 1-RZ | | 17 | 12 | 79 | | |
| Q9 | DTA CLK-S to STB-CS | | 21 | 16 | 94 | | |
| Q10 | DTA CLK-S to DTA 0-RZ | | 23 | 12 | 79 | | |
| Q11 | DTA CLK-S to LRC-S | | 26 | 12 | 71 | | |
| Q12 | DTA CLK-S to DTA 1-S | | 31 | 24 | 143 | | |





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NOTES
1 R1 = R2 = 51 Ω ±1%, 1/4 W
2 S1 = S2 = SPDT switch
3 Load all outputs except pin 38 as specified on figure 16
4 After making all electrical connections the following switching sequence must be performed before making measurement Switch S1 from position 1 to position 2 for a minimum of 100 ns and return it to position 1, then switch S2 from position 2 to position 1 for a minimum of 100 ns and return it to position 2

FIGURE 22 Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit for tables XVI and XVII.



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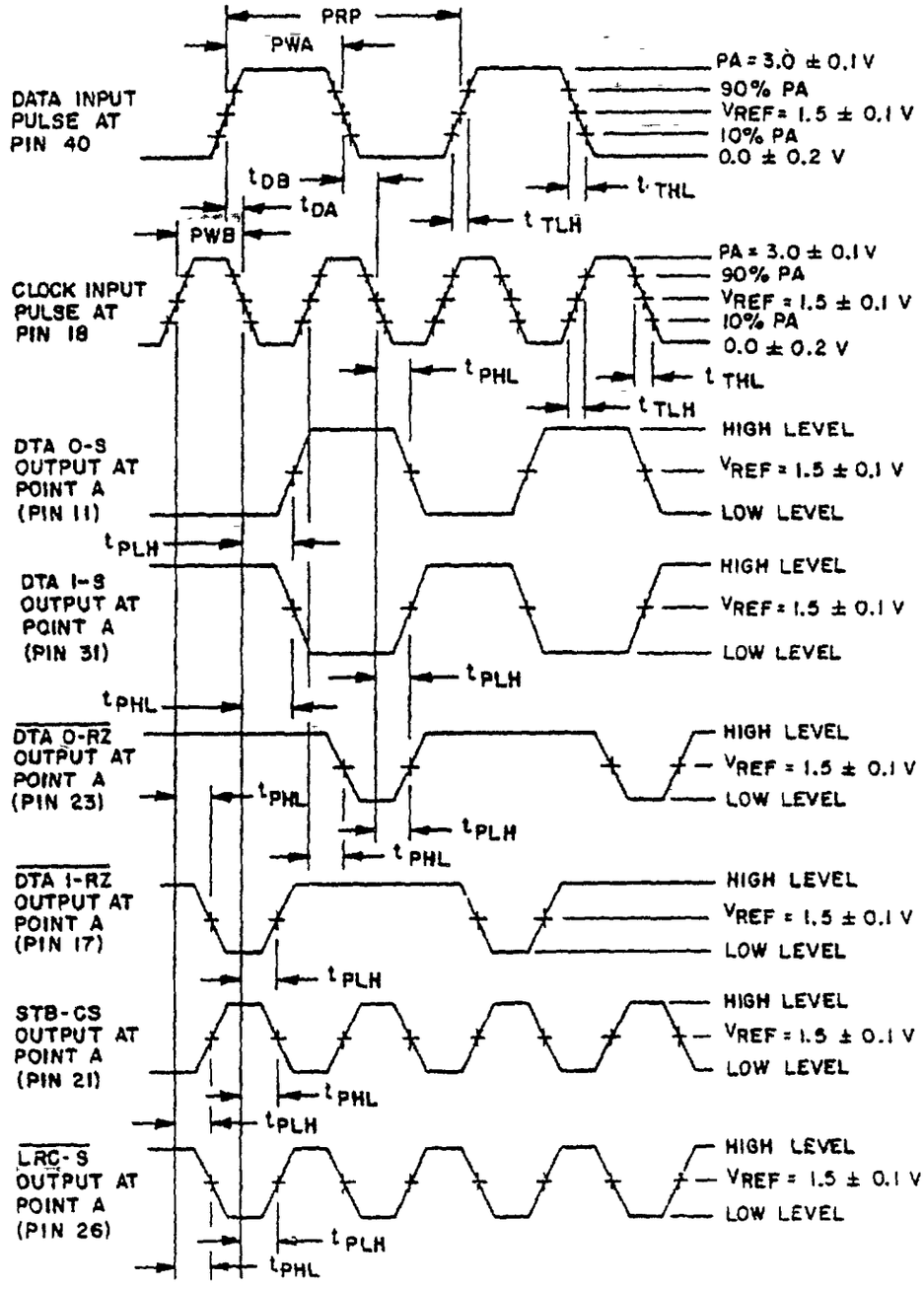



FIGURE 23 Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for tables XVI and XVII



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NOTES:
1. Data input pulse parameters.
   PRP = 1,000 ±100 ns.
   PWA = 500 ±50 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Clock input pulse parameters.
   PWB = 250 ±25 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   $t_{DA} = t_{DB}$ = 85 ±8 ns.
3. The clock pulse generator shall be set in the double pulse mode. Adjust the width control to obtain PWB and the delay control to obtain $t_{DB}$. Adjust the delay control on the data pulse generator to obtain $t_{DA}$.
4. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making any measurements.
5. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes YO1-1 and YO1-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 23. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test circuit waveforms for tables XVI and XVII - Continued



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TABLE XVIII. Skew, spike width, and pulse width test.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| R1 | $\overline{\text{LRC-S}}$ to SRM | Skew | 26 | 16 | -6 | 100 | ns | 24,25,26 |
| R2 | $\overline{\text{DTA 0-RZ}}$ to $\overline{\text{DTA 1-RZ}}$ | Skew | 23 | 17 | -5 | 5 | ↓ | ↓ |
| S1 | $\overline{\text{EOC}}$' | Spike width | 16 | 35 | 0 | 3 | | |
| S2 | $\overline{\text{EOC}}$' | Pulse width | 16 | 35 | 40 | 150 | | |





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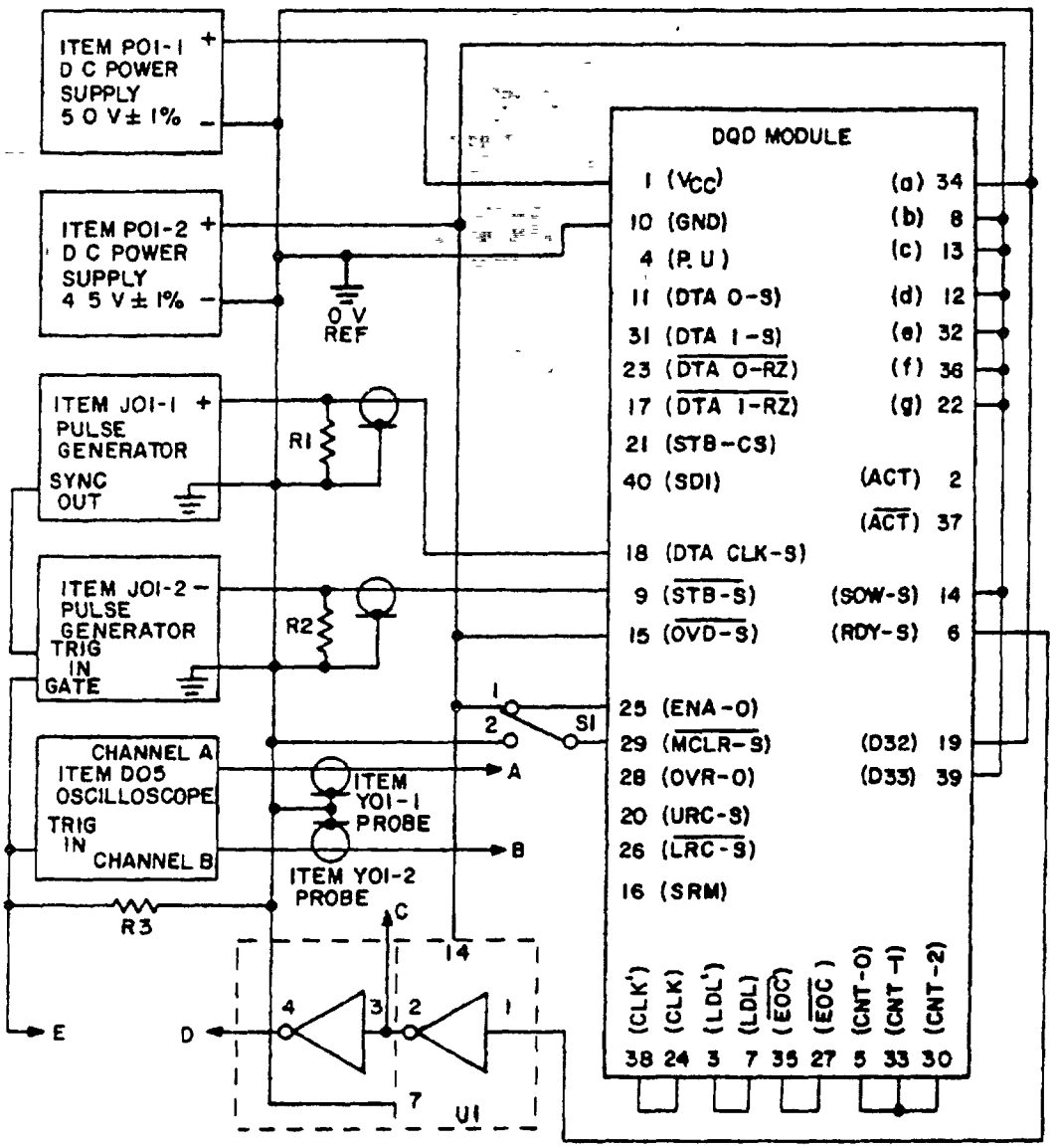



FIGURE 24. Skew, spike width, and pulse width test circuit for table XVIII.



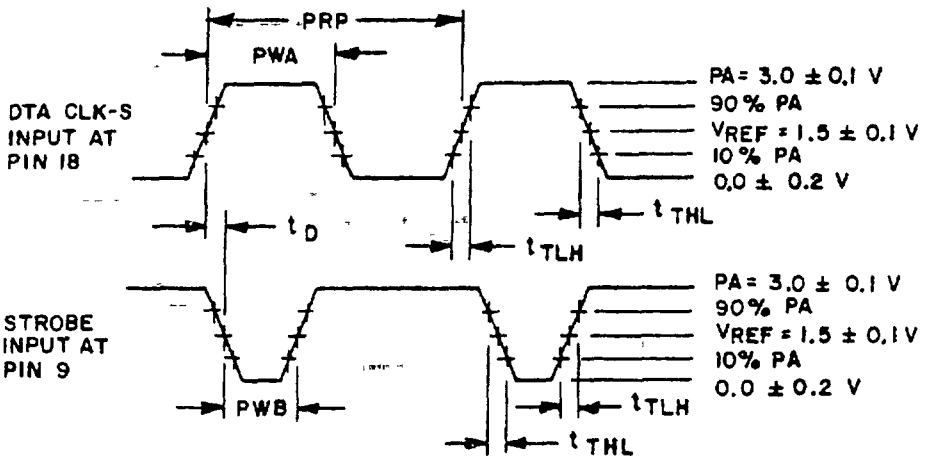
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 402Ω ±1%, 1/4 W.
3. S1 = SPDT switch.
4. U1 = SN5404 Hex Inverter, or equivalent.
5. Load output pins 16, 17, 23 and 26 as specified on figure 16.
6. If item J01-2 is inhibited by a high level on the gate input, connect point E to point C. If item J01-2 is inhibited by a low level on the gate input, connect point E to point D.
7. After making all electrical connections the following switching sequence must be performed before making measurement. Switch S1 from position 1 to position 2 for a minimum of 100 ns and return it to position 1.
8. The following electrical connections apply during adjustment of outputs from items J01-1 and J01-2:
   Pin 2 of U1 = Open.
   Gate connection of item J01-2 = Open.
   Trig in of item D05 = Connected to item J01-1 sync out.

FIGURE 24. Skew, spike width, and pulse width test circuit for table XVIII - Continued.



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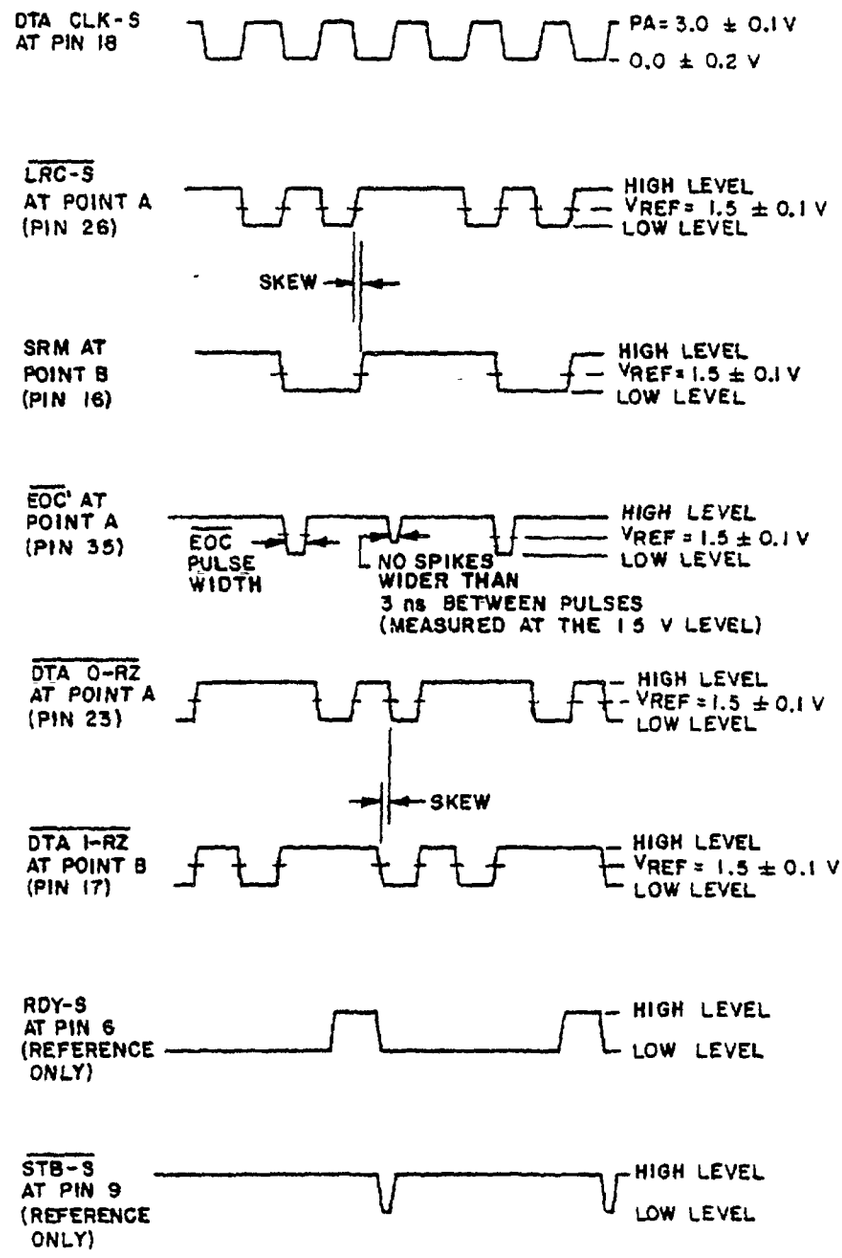



NOTES
1 DTA CLK-S pulse parameters:
PRP = 500 ±50 ns.
PWA = 250 ±25 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns
2 Strobe pulse parameters
PWB = 60 ±6 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.
$t_D$ = 50 ±10 ns.

FIGURE 25. Skew, spike width, and pulse width test circuit waveforms for table XVIII.



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NOTES
1. Verify that all outputs ha e a high level ≥ 2 4 V and a low level ≤ 0 4 V
2 Physical length of electrical conductors between resistors R1 and R2 and input pins; between probes YO1-1 and YO1-2 and nput pins, and between load circuits and output pins, shall be less than 1 inch (25 4 mm)
3 All voltages are referenced to 0 V unless otherwise specified

FIGURE 26. Skew, spike width, and pulse width test circuit waveforms for table VIII.



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TABLE XIX Output crosstalk (high-level) test.

| Test condition | Parameter | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 24 to | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| T1 | CNT-0 | 38 | 27,14 | 7,29 | 9,18, 25 | Open | 5 | 0 | 400 | mV | 27, 28 |
| T2 | RDY-S | | 14,27 | 9,15,29 | 18 | 7,25 | 6 | | | | |
| T3 | DTA 0-S | | 14,19, 39,40 | 29 | 18 | 7,9,25, 27 | 11 | | | | |
| T4 | SRM | | 7,14 | 9,15,29 | 18 | 25,27 | 16 | | | | |
| T5 | $\overline{\text{DTA 1-RZ}}$ | Open | 18 | 15,29 | 24 | 7,9,25, 27 | 17 | | | | |
| T6 | URC-S | 38 | 9 | 15,29 | 18 | 7,25,27 | 20 | | | | |
| T7 | STB-CS | Open | Open | 7,15,18, 25,27,29 | 24 | 9 | 21 | | | | |
| T8 | $\overline{\text{DTA 0-RZ}}$ | Open | 18 | 15,29 | 24 | 7,9,25, 27 | 23 | | | | |
| T9 | $\overline{\text{LRC-S}}$ | Open | 18,27 | 14,15,29 | 24 | 7,9,25 | 26 | | | | |
| T10 | DTA 1-S | 38 | 14 | 15,19,27, 29,39,40 | 18 | 7,9,25 | 31 | | | | |
| T11 | $\overline{\text{EOC}}'$ | 38 | 8,27 | 12,13,15, 22,29,32, 34,36 | 18 | 7,9,25 | 35 | | | | |
| T12 | CLK' | Open | Open | 14,15,18, 29 | 24 | 7,9,25, 27 | 38 | | | | |
| T13 | LDL' | 38 | 7,9,27 | 15,25,29 | 18 | Open | 3 | | | | |
| T14 | ACT | 38 | 27 | 7,15,29 | 18 | 14,25 | 2 | | | | |
| T15 | $\overline{\text{ACT}}$ | 38 | 7,27 | 15,29 | 18 | 14,25 | 37 | | | | |



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TABLE XX. Output crosstalk (low-level) test.

| Test condition | Parameter | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pin 24 to | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| U1 | CNT-O | Open | 14 | 15,18,29 | 24 | 7,9,25 | 5 | 0 | 400 | mV | 27, 28 |
| U2 | RDY-S | 38 | 9 | 14,15,29 | 18 | 7,25,27 | 6 | | | | |
| U3 | DTA 0-S | | 14 | 15,19,29, 39,40 | | 7,9,25, 27 | 11 | | | | |
| U4 | SRM | | Open | 7,15,29 | | 9,14, 25,27 | 16 | | | | |
| U5 | URC-S | | Open | 9,14,15, 29 | | 7,25,27 | 20 | | | | |
| U6 | S.B-CS | Open | 18 | 15,29 | 24 | 7,9,25, 27 | 21 | | | | |
| U7 | $\overline{\text{LRC-S}}$ | Open | 14 | 15,18,27, 29 | 24 | 7,9,25 | 26 | | | | |
| U8 | OVR-O | Open | 18 | 14,15,29 | 24 | 7,9,25, 27 | 28 | | | | |
| U9 | DTA 1-S | 38 | 14,19, 39,40 | 15,29 | 18 | 7,9,25, 27 | 31 | | | | |
| U10 | $\overline{\text{EOC}}$' | 38 | 27 | 8,12,13, 14,15,22, 29,32,34, 36 | 18 | 7,9,25 | 35 | | | | |
| U11 | CLK' | Open | 18 | 14,15 | 24 | 7,25, 27,29 | 38 | | | | |
| U12 | LDI' | 38 | Open | 14,15,27, 29 | 18 | 7,9,25 | 3 | | | | |
| U13 | ACT | 38 | 7,27 | 15,29 | 18 | 25 | 2 | | | | |
| U14 | $\overline{\text{ACT}}$ | 38 | 27 | 7,15,29 | 18 | 14,25 | 37 | | | | |



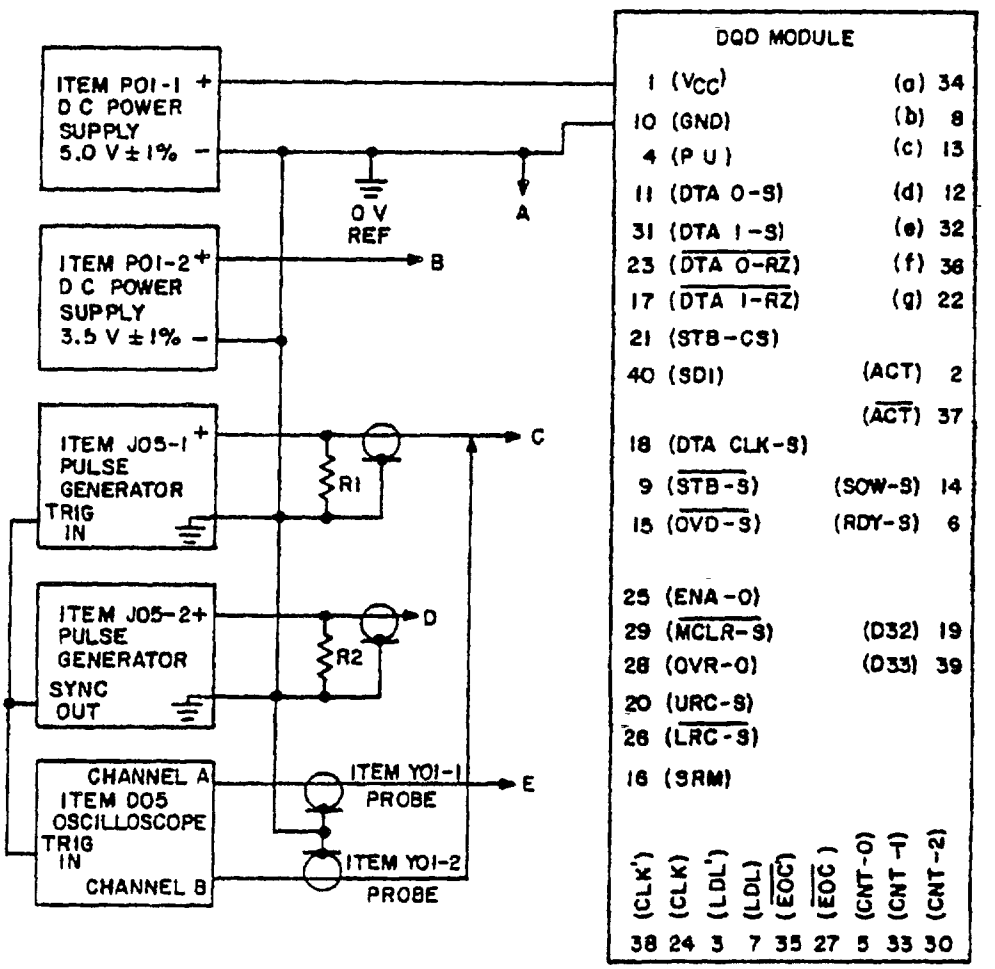
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TABLE XX Output crosstalk (low-level) test - Continued.

| Test condition | Parameter | | Connect | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Pin 24 to | A to pin | B to pin | C to pin | D to pin | E to pin | Min | Max | | |
| U15 | $\overline{DIA\ 0}$-$\overline{RZ}$ | Output crosstalk, low level | Open | Open | 7,14,15, 18,29 | 24 | 9,25,27 | 23 | 0 | 400 | mV | 27, 28 |
| U16 | $\overline{DIA\ 1}$-$\overline{RZ}$ | Output crosstalk, low level | Open | Open | 7,14,15, 18,29 | 24 | 9,25,27 | 17 | 0 | 400 | mV | 27, 28 |



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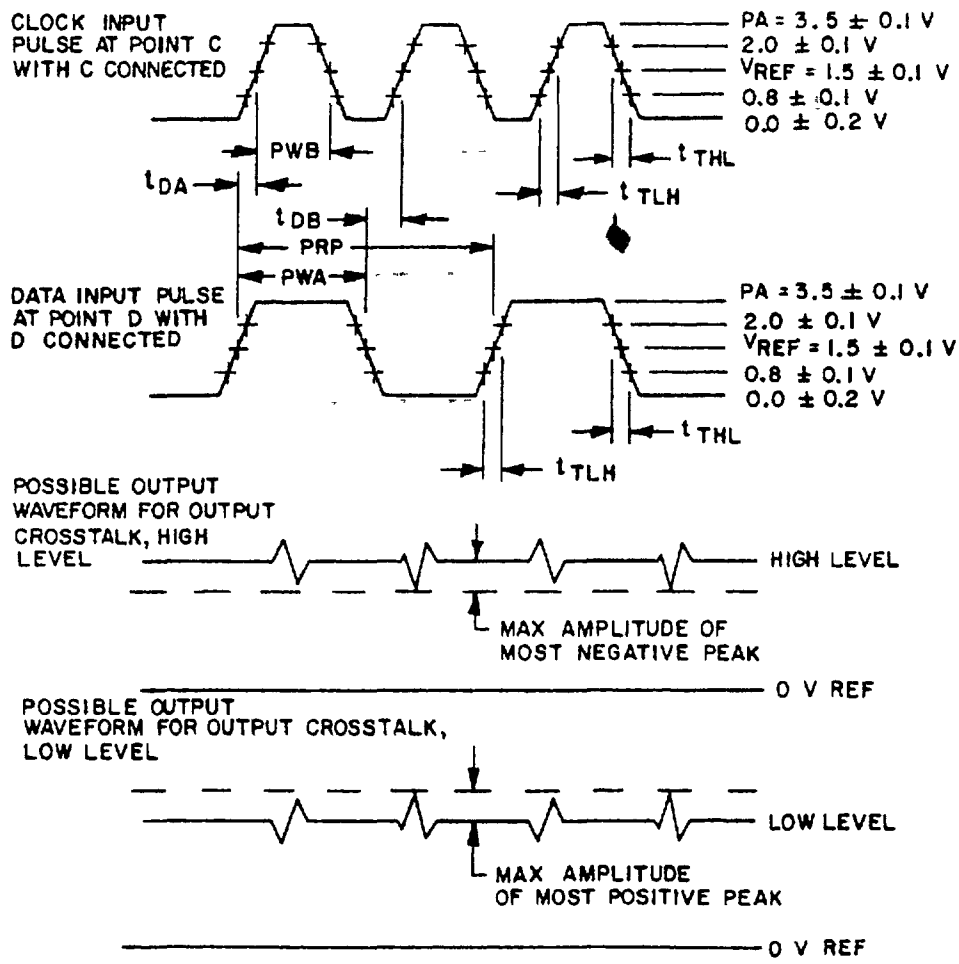



NOTES.
1 R1 = R2 = 51 1Ω ±1%, 1/4 W
2 Load all outputs as specified on figure 16

FIGURE 27 Output crosstalk (high-level and low-level) test circuit for tables XIX and XX



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NOTES

1 Data input pulse parameters
  PRP = 2.0 ±0.2 µs
  PWA = 1.0 ±0.1 µs.
  $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)

2 Clock input pulse parameters
  PWB = 200 ±20 ns.
  $t_{DA}$ = 200 ±20 ns
  $t_{DB}$ = 200 ±20 ns
  $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin)

3 Item J05-1 is run in the double pulse mode. Adjust the delay control to set $t_{DB}$

4 Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm)

5 All voltages are referenced to 0 V unless otherwise specified

FIGURE 28 Output crosstalk (high-level and low-level) test circuit waveforms for tables XIX and XX.



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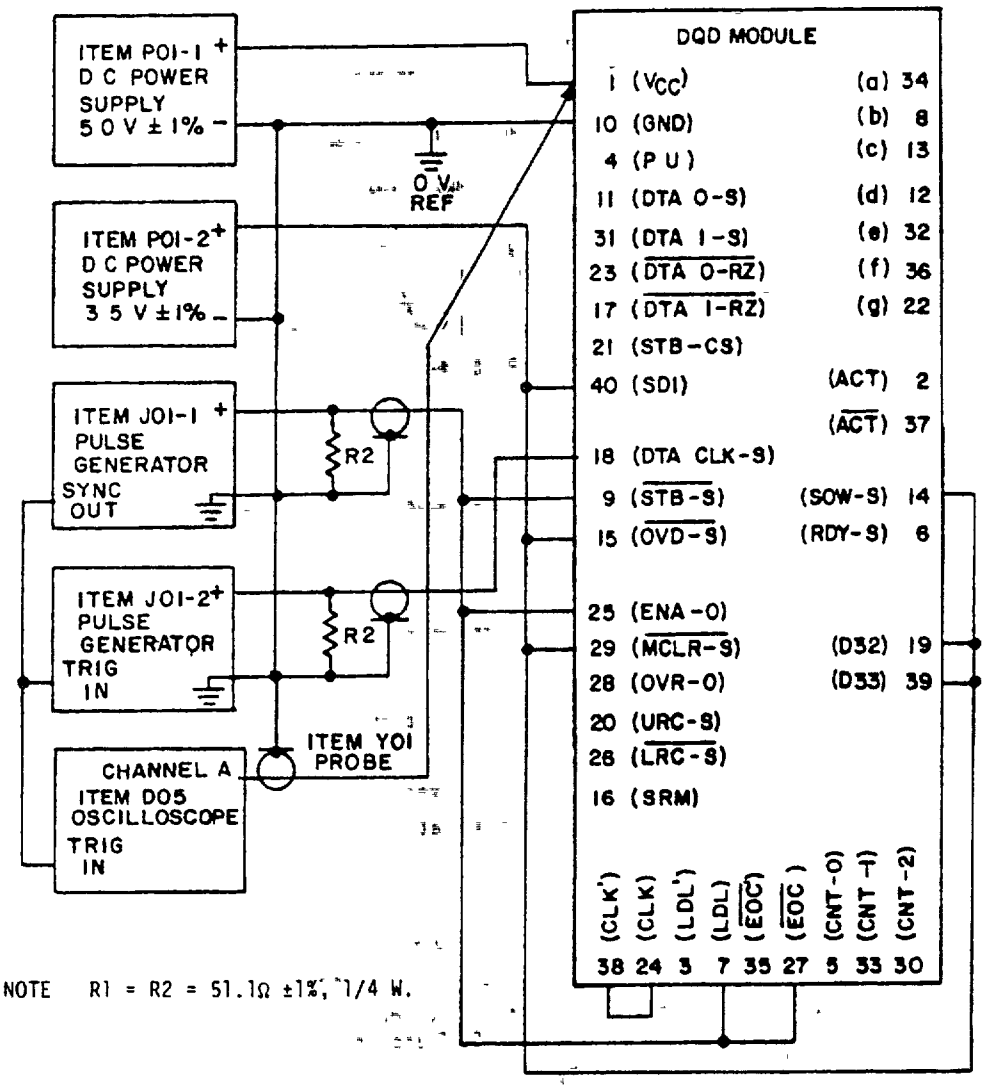
TABLE XXI. Power supply transient amplitude test.

| Test condition | Parameter | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| V | Power supply transient amplitude | 0 | 600 | mV | 29, 30 |





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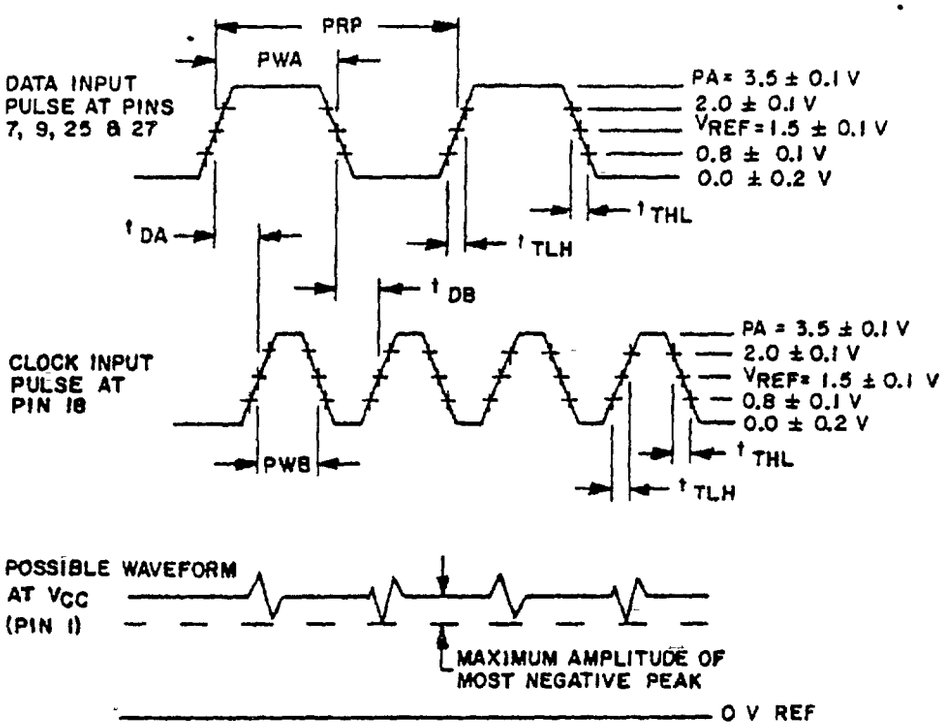



NOTE R1 = R2 = 51.1Ω ±1%, 1/4 W.

FIGURE 29 Power supply transient amplitude test circuit for table XXI



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NOTES:
1. Data input pulse parameters at pins 7, 9, 25, and 27:
   PRP = 2.0 ±0.2 µs.
   PWA = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. Clock input pulse parameters at pin 18:
   PWB = 200 ±20 ns.
   $t_{DA}$ = 200 ±20 ns.
   $t_{DB}$ = 200 ±20 ns
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin)
3. Item J05-1 is run in the double pulse mode. Adjust the delay control to set $t_{DB}$
4. Physical length of electrical conductors between resistors R1 and R2 and input pins, between probes Y01-1 and Y01-2 and input pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
5. All voltages are referenced to 0 V unless otherwise specified

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XII



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TABLE XXII Input capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| W1 | b Input capacitance | 8 | Open | 2 | 15 | pF | 31 |
| W2 | d | 12 | | | | | |
| W3 | c | 13 | | | | | |
| W4 | SOW-S | 14 | | | | | |
| W5 | D32 | 19 | | | | | |
| W6 | g | 22 | | | | | |
| W7 | $\overline{\text{MCLR-S}}$ | 29 | | | | | |
| W8 | CNT-2 | 30 | | | | | |
| W9 | e | 32 | | | | | |
| W10 | CNT-1 | 33 | | | | | |
| W11 | a | 34 | | | | | |
| W12 | f | 36 | | | | | |
| W13 | D33 | 39 | | | | | |
| W14 | $\overline{\text{STB-S}}$ | 9 | | | 25 | | |
| W15 | $\overline{\text{OVD-S}}$ | 15 | | | | | |
| W16 | DTA CLK-S | 18 | | | | | |
| W17 | ENA-0 | 25 | | | | | |
| W18 | $\overline{\text{EOC}}$ | 27 | | | | | |
| W19 | SDI | 40 | | | | | |





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TABLE XXII. Input capacitance test - Continued.

| Test condition | Parameter | | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | Min | Max | | |
| W20 | CLK | Input capacitance | 24 | Open | 2 | 120 | pF | 31 |
| W21 | LDL | Input capacitance | 7 | Open | 2 | 95 | pF | 31 |





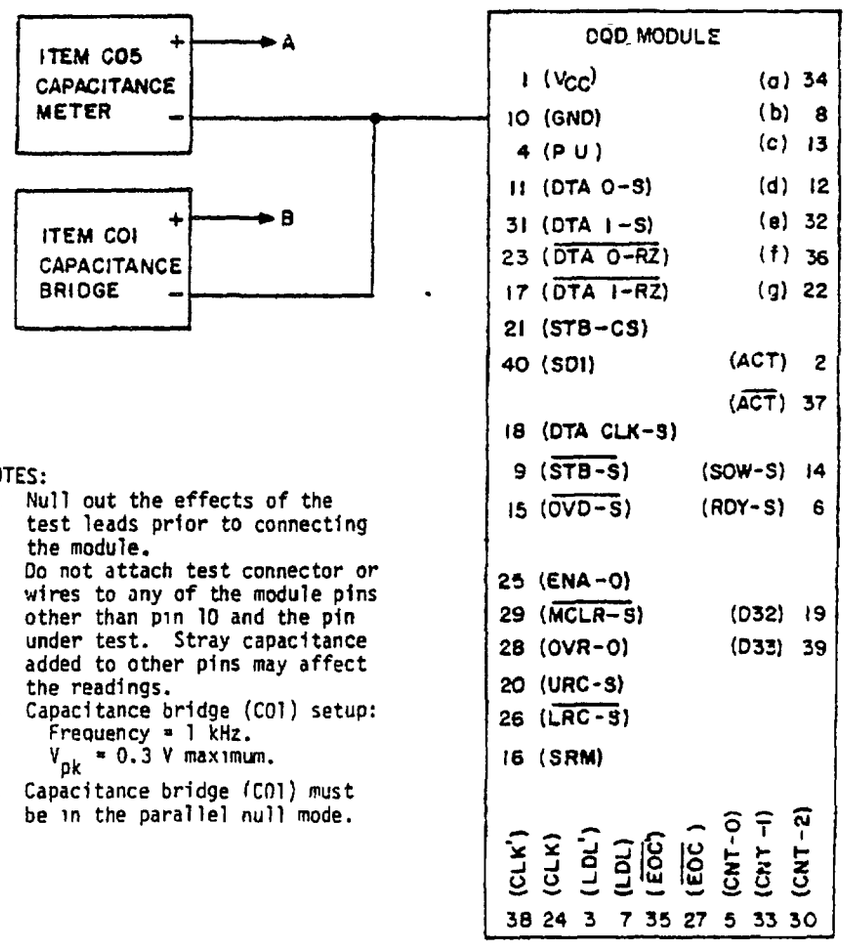
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TABLE XXIII Power supply filter capacitance test.

| Test condition | Parameter | Connect | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| X | Power supply filter capacitance | Open | 1 | 0.07 | 0.13 | µF | 31 |



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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 1 kHz.
   $V_{pk}$ = 0.3 V maximum.
4. Capacitance bridge (C01) must be in the parallel null mode.

FIGURE 31 Input and power supply filter capacitance test circuit for tables XXII and XXIII



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